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Chapter

4

Higher-Order Differential Equations

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Chapter 4 In Review

We turn now to the solution of ordinary differential equations of order two or higher. In the first seven sections of this chapter we examine the underlying theory and solution methods for certain kinds of linear DEs. The chapter concludes with a brief examination of solution methods for nonlinear higher-order equations.

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4.1 Theory of Linear Equations

Reread (*vi*) of the *Remarks* in Section 1.1 and page 57 of Section 2.3.

Introduction In Chapter 2 we were able to solve a few first-order differential equations by recognizing them as separable, linear, exact, or having homogeneous coefficients. Even though the solutions obtained were in the form of a one-parameter family, this family, with one exception, did not represent the *general solution* of the differential equation. Recall that a **general solution** is a family of solutions defined on some interval I that contains *all* solutions of the DE that are defined on I. Only in the case of *linear* first-order differential equations were we able to obtain general solutions by paying close attention to certain continuity conditions imposed on the coefficients in the equation. Because our primary goal in this chapter is to find general solutions of linear higher-order DEs, we first need to examine some of the basic theory of linear equations.

4.1.1 Initial-Value and Boundary-Value Problems

Initial-Value Problem In Section 1.2 we defined an initial-value problem for a general nth-order differential equation. For a linear differential equation an ***n*th-order initial-value problem (IVP)** is

$$\text{Solve:} \quad a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \tag{1}$$

$$\text{Subject to:} \quad y(x_0) = y_0, \quad y'(x_0) = y_1, \ldots, y^{(n-1)}(x_0) = y_{n-1}.$$

Recall that for a problem such as this one we seek a function defined on some interval I, containing x_0, that satisfies the differential equation and the n initial conditions specified at x_0: $y(x_0) = y_0$, $y'(x_0) = y_1$, … , $y^{(n-1)}(x_0) = y_{n-1}$. We have already seen that in the case of a second-order initial-value problem a solution curve must pass through the point (x_0, y_0) and have slope y_1 at this point.

Existence and Uniqueness In Section 1.2 we stated a theorem that gave conditions under which the existence and uniqueness of a solution of a first-order initial-value problem were guaranteed. The theorem that follows gives sufficient conditions for the existence of a unique solution of the problem in (1).

Theorem 4.1.1 Existence of a Unique Solution

Let $a_n(x), a_{n-1}(x), \ldots, a_1(x), a_0(x)$ and $g(x)$ be continuous on an interval I and let $a_n(x) \neq 0$ for every x in this interval. If $x = x_0$ is any point in this interval, then a solution $y(x)$ of the initial-value problem (1) exists on the interval and is unique.

Example 1 Unique Solution of an IVP

The initial-value problem

$$3y''' + 5y'' - y' + 7y = 0, \quad y(1) = 0, \quad y'(1) = 0, \quad y''(1) = 0$$

possesses the trivial solution $y = 0$. Because the third-order equation is linear with constant coefficients, it follows that all the conditions of Theorem 4.1.1 are fulfilled. Hence $y = 0$ is the *only* solution on any interval containing $x = 1$. ■

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Example 2 Unique Solution of an IVP

You should verify that the function $y = 3e^{2x} + e^{-2x} - 3x$ is a solution of the initial-value problem

$$y'' - 4y = 12x, \quad y(0) = 4, \quad y'(0) = 1.$$

Now the differential equation is linear, the coefficients as well as $g(x) = 12x$ are continuous, and $a_2(x) = 1 \neq 0$ on any interval I containing $x = 0$. We conclude from Theorem 4.1.1 that the given function is the unique solution on I. ■

The requirements in Theorem 4.1.1 that $a_i(x)$, $i = 0, 1, 2, \ldots, n$ be continuous and $a_n(x) \neq 0$ for every x in I are both important. Specifically, if $a_n(x) = 0$ for some x in the interval, then the solution of a linear initial-value problem may not be unique or even exist. For example, you should verify that the function $y = cx^2 + x + 3$ is a solution of the initial-value problem

$$x^2y'' - 2xy' + 2y = 6, \quad y(0) = 3, \quad y'(0) = 1$$

on the interval $(-\infty, \infty)$ for any choice of the parameter c. In other words, there is no unique solution of the problem. Although most of the conditions of Theorem 4.1.1 are satisfied, the obvious difficulties are that $a_2(x) = x^2$ is zero at $x = 0$ and that the initial conditions are also imposed at $x = 0$.

Boundary-Value Problem Another type of problem consists of solving a linear differential equation of order two or greater in which the dependent variable y or its derivatives are specified at *different points*. A problem such as

$$\textbf{Solve:} \quad a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

$$\textbf{Subject to:} \quad y(a) = y_0, \quad y(b) = y_1$$

is called a **two-point boundary-value problem,** or simply a **boundary-value problem (BVP).** The prescribed values $y(a) = y_0$ and $y(b) = y_1$ are called **boundary conditions (BC).** A solution of the foregoing problem is a function satisfying the differential equation on some interval I, containing a and b, whose graph passes through the two points (a, y_0) and (b, y_1). See Figure 4.1.1.

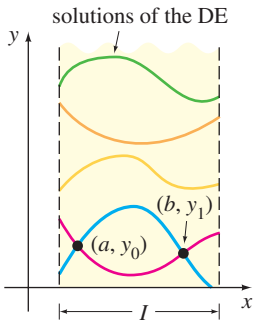


Figure 4.1.1 Solution curves of a BVP that pass through two points

For a second-order differential equation other pairs of boundary conditions could be

$$\begin{aligned} y'(a) &= y_0, & y(b) &= y_1 \\ y(a) &= y_0, & y'(b) &= y_1 \\ y'(a) &= y_0, & y'(b) &= y_1, \end{aligned}$$

where y_0 and y_1 denote arbitrary constants. These three pairs of conditions are just special cases of the general boundary conditions

$$\begin{aligned} \alpha_1 y(a) + \beta_1 y'(a) &= \gamma_1 \\ \alpha_2 y(b) + \beta_2 y'(b) &= \gamma_2. \end{aligned}$$

The next example shows that even when the conditions of Theorem 4.1.1 are fulfilled, a boundary-value problem may have several solutions (as suggested in Figure 4.1.1), a unique solution, or no solution at all.

Example 3 A BVP Can Have Many, One, or No Solutions

In Example 9 of Section 1.1 we saw that the two-parameter family of solutions of the differential equation $x'' + 16x = 0$ is

$$x = c_1 \cos 4t + c_2 \sin 4t. \tag{2}$$

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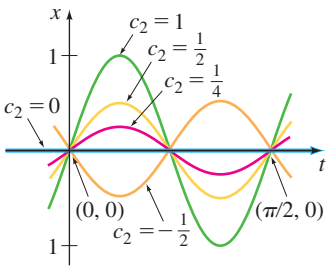


Figure 4.1.2 Solution curves for BVP in part (a) of Example 3

(a) Suppose we now wish to determine the solution of the equation that further satisfies the boundary conditions $x(0) = 0$, $x(\pi/2) = 0$. Observe that the first condition $0 = c_1 \cos 0 + c_2 \sin 0$ implies that $c_1 = 0$, so $x = c_2 \sin 4t$. But when $t = \pi/2$, $0 = c_2 \sin 2\pi$ is satisfied for any choice of c_2, since $\sin 2\pi = 0$. Hence the boundary-value problem

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 0 \tag{3}$$

has infinitely many solutions. Figure 4.1.2 shows the graphs of some of the members of the one-parameter family $x = c_2 \sin 4t$ that pass through the two points (0, 0) and $(\pi/2, 0)$.

(b) If the boundary-value problem in (3) is changed to

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{8}\right) = 0, \tag{4}$$

then $x(0) = 0$ still requires $c_1 = 0$ in the solution (2). But applying $x(\pi/8) = 0$ to $x = c_2 \sin 4t$ demands that $0 = c_2 \sin(\pi/2) = c_2 \cdot 1$. Hence $x = 0$ is a solution of this new boundary-value problem. Indeed, it can be proved that $x = 0$ is the *only* solution of (4).

(c) Finally, if we change the problem to

$$x'' + 16x = 0, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 1, \tag{5}$$

we find again from $x(0) = 0$ that $c_1 = 0$, but applying $x(\pi/2) = 1$ to $x = c_2 \sin 4t$ leads to the contradiction $1 = c_2 \sin 2\pi = c_2 \cdot 0 = 0$. Hence the boundary-value problem (5) has no solution. ■

4.1.2 Homogeneous Equations

A linear nth-order differential equation of the form

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0 \tag{6}$$

is said to be **homogeneous,** whereas an equation

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x), \tag{7}$$

with $g(x)$ not identically zero, is said to be **nonhomogeneous.** For example, $2y'' + 3y' - 5y = 0$ is a homogeneous linear second-order differential equation, whereas $x^3y''' + 6y' + 10y = e^x$ is a nonhomogeneous linear third-order differential equation. The word *homogeneous* in this context does not refer to coefficients that are homogeneous functions, as in Section 2.5.

Note $y = 0$ is always a solution of a homogeneous linear equation.

We shall see that to solve a nonhomogeneous linear equation (7), we must first be able to solve the **associated homogeneous equation** (6).

To avoid needless repetition throughout the remainder of this text, we shall, as a matter of course, make the following important assumptions when stating definitions and theorems about linear equations (1). On some common interval I,

Please remember these two assumptions.

- the coefficient functions $a_i(x)$, $i = 0, 1, 2, \ldots, n$ and $g(x)$ are continuous;
- $a_n(x) \neq 0$ for every x in the interval.

Differential Operators In calculus differentiation is often denoted by the capital letter D—that is, $dy/dx = Dy$. The symbol D is called a **differential operator** because it transforms a differentiable function into another function. For example,

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$D(\cos 4x) = -4 \sin 4x$ and $D(5x^3 - 6x^2) = 15x^2 - 12x$. Higher-order derivatives can be expressed in terms of D in a natural manner:

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = D(Dy) = D^2y \qquad \text{and, in general,} \qquad \frac{d^ny}{dx^n} = D^ny,$$

where y represents a sufficiently differentiable function. Polynomial expressions involving D, such as $D + 3$, $D^2 + 3D - 4$, and $5x^3D^3 - 6x^2D^2 + 4xD + 9$, are also differential operators. In general, we define an ***n*th-order differential operator** or **polynomial operator** to be

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \cdots + a_1(x)D + a_0(x). \tag{8}$$

As a consequence of two basic properties of differentiation, $D(cf(x)) = cDf(x)$, c is a constant, and $D\{f(x) + g(x)\} = Df(x) + Dg(x)$, the differential operator L possesses a linearity property; that is, L operating on a linear combination of two differentiable functions is the same as the linear combination of L operating on the individual functions. In symbols this means that

$$L\{\alpha f(x) + \beta g(x)\} = \alpha L(f(x)) + \beta L(g(x)), \tag{9}$$

where α and β are constants. Because of (9) we say that the nth-order differential operator L is a **linear operator.**

Differential Equations Any linear differential equation can be expressed in terms of the D notation. For example, the differential equation $y'' + 5y' + 6y = 5x - 3$ can be written as $D^2y + 5Dy + 6y = 5x - 3$ or $(D^2 + 5D + 6)y = 5x - 3$. Using (8), we can write the linear nth-order differential equations (6) and (7) compactly as

$$L(y) = 0 \qquad \text{and} \qquad L(y) = g(x),$$

respectively.

Superposition Principle In the next theorem we see that the sum, or **superposition,** of two or more solutions of a homogeneous linear differential equation is also a solution.

Theorem 4.1.2 Superposition Principle—Homogeneous Equations

Let $y_1, y_2, \ldots, y_k$ be solutions of the homogeneous nth-order differential equation (6) on an interval I. Then the linear combination

$$y = c_1y_1(x) + c_2y_2(x) + \cdots + c_ky_k(x),$$

where the c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution on the interval.

Proof We prove the case $k = 2$. Let L be the differential operator defined in (8), and let $y_1(x)$ and $y_2(x)$ be solutions of the homogeneous equation $L(y) = 0$. If we define $y = c_1y_1(x) + c_2y_2(x)$, then by linearity of L we have

$$L(y) = L\{c_1y_1(x) + c_2y_2(x)\} = c_1\,L(y_1) + c_2\,L(y_2) = c_1 \cdot 0 + c_2 \cdot 0 = 0. \quad \blacksquare$$

Corollaries to Theorem 4.1.2

(A) A constant multiple $y = c_1y_1(x)$ of a solution $y_1(x)$ of a homogeneous linear differential equation is also a solution.

(B) A homogeneous linear differential equation always possesses the trivial solution $y = 0$.

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Example 4 Superposition—Homogeneous DE

The functions $y_1 = x^2$ and $y_2 = x^2 \ln x$ are both solutions of the homogeneous linear equation $x^3y''' - 2xy' + 4y = 0$ on the interval $(0, \infty)$. By the superposition principle the linear combination

$$y = c_1x^2 + c_2x^2 \ln x$$

is also a solution of the equation on the interval. ■

The function $y = e^{7x}$ is a solution of $y'' - 9y' + 14y = 0$. Because the differential equation is linear and homogeneous, the constant multiple $y = ce^{7x}$ is also a solution. For various values of c we see that $y = 9e^{7x}$, $y = 0$, $y = -\sqrt{5}e^{7x}$, . . . are all solutions of the equation.

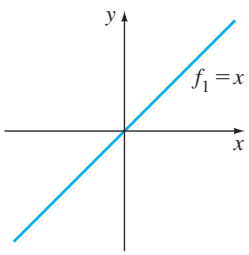
Linear Dependence and Linear Independence The next two concepts are basic to the study of linear differential equations.

Definition 4.1.1 Linear Dependence/Independence

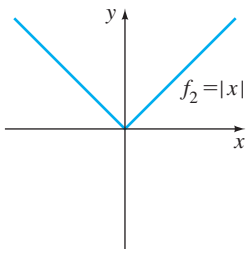
A set of functions $f_1(x), f_2(x), \ldots, f_n(x)$ is said to be **linearly dependent** on an interval I if there exist constants $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1f_1(x) + c_2f_2(x) + \cdots + c_nf_n(x) = 0$$

for every x in the interval. If the set of functions is not linearly dependent on the interval, it is said to be **linearly independent.**



(a)



(b)

Figure 4.1.3 Set consisting of f_1 and f_2 is linearly independent on $(-\infty, \infty)$

In other words, a set of functions is linearly independent on an interval I if the only constants for which

$$c_1f_1(x) + c_2f_2(x) + \cdots + c_nf_n(x) = 0$$

for every x in the interval are $c_1 = c_2 = \cdots = c_n = 0$.

It is easy to understand these definitions for a set consisting of two functions $f_1(x)$ and $f_2(x)$. If the set of functions is linearly dependent on an interval, then there exist constants c_1 and c_2 that are not both zero such that for every x in the interval, $c_1f_1(x) + c_2f_2(x) = 0$. Therefore if we assume that $c_1 \neq 0$, it follows that $f_1(x) = (-c_2/c_1)f_2(x)$; that is, *if a set of two functions is linearly dependent, then one function is simply a constant multiple of the other.* Conversely, if $f_1(x) = c_2f_2(x)$ for some constant c_2, then $(-1) \cdot f_1(x) + c_2f_2(x) = 0$ for every x in the interval. Hence, the set of functions is linearly dependent because at least one of the constants (namely, $c_1 = -1$) is not zero. We conclude that *a set of two functions* $f_1(x)$ and $f_2(x)$ *is linearly independent when neither function is a constant multiple of the other* on the interval. For example, the set of functions $f_1(x) = \sin 2x$, $f_2(x) = \sin x \cos x$ is linearly dependent on $(-\infty, \infty)$ because $f_1(x)$ is a constant multiple of $f_2(x)$. Recall from the double-angle formula for the sine that $\sin 2x = 2 \sin x \cos x$. On the other hand, the set of functions $f_1(x) = x$, $f_2(x) = |x|$ is linearly independent on $(-\infty, \infty)$. Inspection of Figure 4.1.3 should convince you that neither function is a constant multiple of the other on the interval.

It follows from the preceding discussion that the quotient $f_2(x)/f_1(x)$ is not a constant on an interval on which the set $f_1(x), f_2(x)$ is linearly independent. This little fact will be used in the next section.

Example 5 Linearly Dependent Set of Functions

The set of functions $f_1(x) = \cos^2x$, $f_2(x) = \sin^2x$, $f_3(x) = \sec^2x$, $f_4(x) = \tan^2x$ is linearly dependent on the interval $(-\pi/2, \pi/2)$ because

$$c_1 \cos^2x + c_2 \sin^2x + c_3 \sec^2x + c_4 \tan^2x = 0$$

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for every real number x in the interval when $c_1 = c_2 = 1$, $c_3 = -1$, $c_4 = 1$. We used here $\cos^2 x + \sin^2 x = 1$ and $1 + \tan^2 x = \sec^2 x$. ■

A set of n functions $f_1(x), f_2(x), \ldots, f_n(x)$ is linearly dependent on an interval I if at least one of the functions can be expressed as a linear combination of the remaining functions. For example, three functions $f_1(x), f_2(x)$, and $f_3(x)$ are linearly dependent on I if at least one of these functions is a linear combination of the other two, say,

$$f_3(x) = c_1 f_1(x) + c_2 f_2(x)$$

for all x in I. A set of n functions is linearly independent on I if no one function is a linear combination of the other functions.

Example 6 Linearly Dependent Set of Functions

The set of functions $f_1(x) = \sqrt{x} + 5, f_2(x) = \sqrt{x} + 5x, f_3(x) = x - 1, f_4(x) = x^2$ is linearly dependent on the interval $(0, \infty)$ because f_2 can be written as a linear combination of f_1, f_3, and f_4. Observe that

$$f_2(x) = 1 \cdot f_1(x) + 5 \cdot f_3(x) + 0 \cdot f_4(x)$$

for every x in the interval $(0, \infty)$. ■

Solutions of Differential Equations We are primarily interested in linearly independent functions or, more to the point, linearly independent *solutions* of a linear differential equation. Although we could always appeal directly to Definition 4.1.1, it turns out that the question of whether the set of n solutions $y_1, y_2, \ldots, y_n$ of a homogeneous linear nth-order differential equation (6) is linearly independent can be settled somewhat mechanically by using a determinant.

Definition 4.1.2 Wronskian

Suppose each of the functions $f_1(x), f_2(x), \ldots, f_n(x)$ possesses at least $n - 1$ derivatives. The determinant

$$W(f_1, f_2, \ldots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix},$$

where the primes denote derivatives, is called the **Wronskian** of the functions.

The Wronskian determinant is named after the Polish philosopher, inventor, lawyer, physicist, and mathematician **Józef Maria Hoëné-Wroński** (1776–1853).

Theorem 4.1.3 Criterion for Linearly Independent Solutions

Let $y_1, y_2, \ldots, y_n$ be n solutions of the homogeneous linear nth-order differential equation (6) on an interval I. Then the set of solutions is **linearly independent** on I if and only if $W(y_1, y_2, \ldots, y_n) \neq 0$ for every x in the interval.

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It follows from Theorem 4.1.3 that when $y_1, y_2, \ldots, y_n$ are n solutions of (6) on an interval I, the Wronskian $W(y_1, y_2, \ldots, y_n)$ is either identically zero or never zero on the interval. Thus, if we can show that $W(y_1, y_2, \ldots, y_n) \neq 0$ for *some* x_0 in I, then the solutions $y_1, y_2, \ldots, y_n$ are linearly independent on I. For example, the functions

$$y_1(x) = \frac{\cos(2\ln x)}{x^3}, \quad y_2(x) = \frac{\sin(2\ln x)}{x^3}$$

and solutions of the differential equation

$$x^2y'' + 7xy' + 13y = 0$$

on the interval $(0, \infty)$. Note that the coefficient functions $a_2(x) = x^2$, $a_1(x) = 7x$, and $a_0(x) = 13$ are continuous on $(0, \infty)$ and that $a_2(x) \neq 0$ for every value of x in the interval. The Wronskian is

$$W(y_1(x), y_2(x)) = \begin{vmatrix} \dfrac{\cos(2\ln x)}{x^3} & \dfrac{\sin(2\ln x)}{x^3} \\ \dfrac{-2x^2\sin(2\ln x) - 3x^2\cos(2\ln x)}{x^6} & \dfrac{2x^2\cos(2\ln x) - 3x^2\sin(2\ln x)}{x^6} \end{vmatrix}.$$

Rather than expanding this unwieldy determinant, we choose $x = 1$ in the interval $(0, \infty)$ and find

$$W(y_1(1), y_2(1)) = \begin{vmatrix} 1 & 0 \\ -3 & 2 \end{vmatrix} = 2.$$

The fact that $W(y_1(1), y_2(1)) = 2 \neq 0$ is sufficient to conclude that $y_1(x)$ and $y_2(x)$ are linearly independent on $(0, \infty)$.

A set of n linearly independent solutions of a homogeneous linear nth-order differential equation is given a special name.

Definition 4.1.3 Fundamental Set of Solutions

Any set $y_1, y_2, \ldots, y_n$ of n linearly independent solutions of the homogeneous linear nth-order differential equation (6) on an interval I is said to be a **fundamental set of solutions** on the interval.

The basic question of whether a fundamental set of solutions exists for a linear equation is answered in the next theorem.

Theorem 4.1.4 Existence of a Fundamental Set

There exists a fundamental set of solutions for the homogeneous linear nth-order differential equation (6) on an interval I.

Analogous to the fact that any vector in three dimensions can be expressed as a linear combination of the *linearly independent* vectors **i, j, k,** any solution of an nth-order homogeneous linear differential equation on an interval I can be expressed as a linear combination of n linearly independent solutions on I. In other words, n linearly independent solutions $y_1, y_2, \ldots, y_n$ are the basic building blocks for the general solution of the equation.

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Theorem 4.1.5 General Solution—Homogeneous Equations

Let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of the homogeneous linear nth-order differential equation (6) on an interval I. Then the **general solution** of the equation on the interval is

$$y = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x),$$

where c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Theorem 4.1.5 states that if $Y(x)$ is any solution of (6) on the interval, then constants $C_1, C_2, \ldots, C_n$ can always be found so that

$$Y(x) = C_1y_1(x) + C_2y_2(x) + \cdots + C_ny_n(x).$$

We will prove the case when $n = 2$.

Proof Let Y be a solution and let y_1 and y_2 be linearly independent solutions of $a_2y'' + a_1y' + a_0y = 0$ on an interval I. Suppose that $x = t$ is a point in I for which $W(y_1(t), y_2(t)) \neq 0$. Suppose also that $Y(t) = k_1$ and $Y'(t) = k_2$. If we now examine the equations

$$C_1y_1(t) + C_2y_2(t) = k_1$$
$$C_1y_1'(t) + C_2y_2'(t) = k_2,$$

it follows that we can determine C_1 and C_2 uniquely, provided that the determinant of the coefficients satisfies

$$\begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} \neq 0.$$

But this determinant is simply the Wronskian evaluated at $x = t$, and by assumption, $W \neq 0$. If we define $G(x) = C_1y_1(x) + C_2y_2(x)$, we observe that $G(x)$ satisfies the differential equation since it is a superposition of two known solutions; $G(x)$ satisfies the initial conditions

$$G(t) = C_1y_1(t) + C_2y_2(t) = k_1 \quad \text{and} \quad G'(t) = C_1y_1'(t) + C_2y_2'(t) = k_2;$$

and $Y(x)$ satisfies the *same* linear equation and the *same* initial conditions. Because the solution of this linear initial-value problem is unique (Theorem 4.1.1), we have $Y(x) = G(x)$ or $Y(x) = C_1y_1(x) + C_2y_2(x)$. ■

Example 7 General Solution of a Homogeneous DE

The functions $y_1 = e^{3x}$ and $y_2 = e^{-3x}$ are both solutions of the homogeneous linear equation $y'' - 9y = 0$ on the interval $(-\infty, \infty)$. By inspection the solutions are linearly independent on the x-axis. This fact can be corroborated by observing that the Wronskian

$$W(e^{3x}, e^{-3x}) = \begin{vmatrix} e^{3x} & e^{-3x} \\ 3e^{3x} & -3e^{-3x} \end{vmatrix} = -6 \neq 0$$

for every x. We conclude that y_1 and y_2 form a fundamental set of solutions, and consequently, $y = c_1e^{3x} + c_2e^{-3x}$ is the general solution of the equation on the interval. ■

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Example 8 A Solution Obtained from a General Solution

The function $y = 4\sinh 3x - 5e^{-3x}$ is a solution of the differential equation in Example 7. (Verify this.) In view of Theorem 4.1.5 we must be able to obtain this solution from the general solution $y = c_1e^{3x} + c_2e^{-3x}$. Observe that if we choose $c_1 = 2$ and $c_2 = -7$, then $y = 2e^{3x} - 7e^{-3x}$ can be rewritten as

$$y = 2e^{3x} - 2e^{-3x} - 5e^{-3x} = 4\left(\frac{e^{3x} - e^{-3x}}{2}\right) - 5e^{-3x}.$$

The last expression is recognized as $y = 4\sinh 3x - 5e^{-3x}$. ■

Example 9 General Solution of a Homogeneous DE

The functions $y_1 = e^x$, $y_2 = e^{2x}$, and $y_3 = e^{3x}$ satisfy the third-order equation $y''' - 6y'' + 11y' - 6y = 0$. Since

$$W(e^x, e^{2x}, e^{3x}) = \begin{vmatrix} e^x & e^{2x} & e^{3x} \\ e^x & 2e^{2x} & 3e^{3x} \\ e^x & 4e^{2x} & 9e^{3x} \end{vmatrix} = 2e^{6x} \neq 0$$

for every real value of x, the functions y_1, y_2, and y_3 form a fundamental set of solutions on $(-\infty, \infty)$. We conclude that $y = c_1e^x + c_2e^{2x} + c_3e^{3x}$ is the general solution of the differential equation on the interval. ■

4.1.3 Nonhomogeneous Equations

Any function y_p, free of arbitrary parameters, that satisfies (7) is said to be a **particular solution** of the equation. For example, it is a straightforward task to show that the constant function $y_p = 3$ is a particular solution of the nonhomogeneous equation $y'' + 9y = 27$.

Now if $y_1, y_2, \ldots, y_k$ are solutions of (6) on an interval I and y_p is any particular solution of (7) on I, then the linear combination

$$y = c_1y_1(x) + c_2y_2(x) + \cdots + c_ky_k(x) + y_p(x) \tag{10}$$

is also a solution of the nonhomogeneous equation (7). If you think about it, this makes sense, because the linear combination $c_1y_1(x) + c_2y_2(x) + \cdots + c_ky_k(x)$ is transformed into 0 by the operator $L = a_nD^n + a_{n-1}D^{n-1} + \cdots + a_1D + a_0$, whereas y_p is transformed into $g(x)$. If we use $k = n$ linearly independent solutions of the nth-order equation (6), then the expression in (10) becomes the general solution of (7).

Theorem 4.1.6 General Solution—Nonhomogeneous Equations

Let y_p be any particular solution of the nonhomogeneous linear nth-order differential equation (7) on an interval I, and let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of the associated homogeneous differential equation (6) on I. Then the **general solution** of the equation on the interval is

$$y = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x) + y_p(x),$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Proof Let L be the differential operator defined in (8) and let $Y(x)$ and $y_p(x)$ be particular solutions of the nonhomogeneous equation $L(y) = g(x)$. If we define $u(x) = Y(x) - y_p(x)$, then by linearity of L we have

$$L(u) = L\{Y(x) - y_p(x)\} = L(Y(x)) - L(y_p(x)) = g(x) - g(x) = 0.$$

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This shows that $u(x)$ is a solution of the homogeneous equation $L(y) = 0$. Hence by Theorem 4.1.5, $u(x) = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x)$, and so

$$Y(x) - y_p(x) = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x)$$

or
$$Y(x) = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x) + y_p(x).$$

Complementary Function We see in Theorem 4.1.6 that the general solution of a nonhomogeneous linear equation consists of the sum of two functions:

$$y = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x) + y_p(x) = y_c(x) + y_p(x).$$

The linear combination $y_c(x) = c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x)$, which is the general solution of (6), is called the **complementary function** for equation (7). In other words, to solve a nonhomogeneous linear differential equation, we first solve the associated homogeneous equation and then find any particular solution of the nonhomogeneous equation. The general solution of the nonhomogeneous equation is then

$$\begin{aligned} y &= \textit{complementary function} + \textit{any particular solution} \\ &= y_c + y_p. \end{aligned}$$

Example 10 General Solution of a Nonhomogeneous DE

By substitution the function $y_p = -\frac{11}{12} - \frac{1}{2}x$ is readily shown to be a particular solution of the nonhomogeneous equation

$$y''' - 6y'' + 11y' - 6y = 3x. \tag{11}$$

To write the general solution of (11), we must also be able to solve the associated homogeneous equation

$$y''' - 6y'' + 11y' - 6y = 0.$$

But in Example 9 we saw that the general solution of this latter equation on the interval $(-\infty, \infty)$ was $y_c = c_1e^x + c_2e^{2x} + c_3e^{3x}$. Hence the general solution of (11) on the interval is

$$y = y_c + y_p = c_1e^x + c_2e^{2x} + c_3e^{3x} - \frac{11}{12} - \frac{1}{2}x.$$

Another Superposition Principle The last theorem of this discussion will be useful in Section 4.4 when we consider a method for finding particular solutions of nonhomogeneous equations.

Theorem 4.1.7 Superposition Principle—Nonhomogeneous Equations

Let $y_{p_1}, y_{p_2}, \ldots, y_{p_k}$ be k particular solutions of the nonhomogeneous linear nth-order differential equation (7) on an interval I corresponding, in turn, to k distinct functions $g_1, g_2, \ldots, g_k$. That is, suppose y_{p_i} denotes a particular solution of the corresponding differential equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_i(x), \tag{12}$$

where $i = 1, 2, \ldots, k$. Then

$$y_p(x) = y_{p_1}(x) + y_{p_2}(x) + \cdots + y_{p_k}(x) \tag{13}$$

is a particular solution of

$$\begin{aligned} &a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y \\ &\quad = g_1(x) + g_2(x) + \cdots + g_k(x). \end{aligned} \tag{14}$$

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Proof We prove the case $k = 2$. Let L be the differential operator defined in (8) and let $y_{p_1}(x)$ and $y_{p_2}(x)$ be particular solutions of the nonhomogeneous equations $L(y) = g_1(x)$ and $L(y) = g_2(x)$, respectively. If we define $y_p(x) = y_{p_1}(x) + y_{p_2}(x)$, we want to show that y_p is a particular solution of $L(y) = g_1(x) + g_2(x)$. The result follows again by the linearity of the operator L:

$$L(y_p) = L\{y_{p_1}(x) + y_{p_2}(x)\} = L(y_{p_1}(x)) + L(y_{p_2}(x)) = g_1(x) + g_2(x).$$ ■

Example 11 Superposition—Nonhomogeneous DE

You should verify that

$$y_{p_1} = -4x^2 \quad \text{is a particular solution of} \quad y'' - 3y' + 4y = -16x^2 + 24x - 8,$$
$$y_{p_2} = e^{2x} \quad \text{is a particular solution of} \quad y'' - 3y' + 4y = 2e^{2x},$$
$$y_{p_3} = xe^x \quad \text{is a particular solution of} \quad y'' - 3y' + 4y = 2xe^x - e^x.$$

It follows from (13) of Theorem 4.1.7 that the superposition of y_{p_1}, y_{p_2}, and y_{p_3},

$$y = y_{p_1} + y_{p_2} + y_{p_3} = -4x^2 + e^{2x} + xe^x,$$

is a solution of

$$y'' - 3y' + 4y = \underbrace{-16x^2 + 24x - 8}_{g_1(x)} + \underbrace{2e^{2x}}_{g_2(x)} + \underbrace{2xe^x - e^x}_{g_3(x)}.$$ ■

Note If the y_{p_i} are particular solutions of (12) for $i = 1, 2, \ldots, k$, then the linear combination

$$y_p = c_1 y_{p_1} + c_2 y_{p_2} + \cdots + c_k y_{p_k},$$

where the c_i are constants, is also a particular solution of (14) when the right-hand member of the equation is the linear combination

$$c_1 g_1(x) + c_2 g_2(x) + \cdots + c_k g_k(x).$$

Before we actually start solving homogeneous and nonhomogeneous linear differential equations, we need one additional bit of theory, which is presented in the next section.

Remarks

This remark is a continuation of the brief discussion of dynamical systems given at the end of Section 1.3.

A dynamical system whose rule or mathematical model is a linear nth-order differential equation

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t)$$

is said to be an nth-order **linear system.** The n time-dependent functions $y(t), y'(t), \ldots, y^{(n-1)}(t)$ are the **state variables** of the system. Recall that their values at some time t give the **state of the system.** The function g is called the **input function** or **forcing function.** A solution $y(t)$ of the differential equation is said to be the **output** or **response** of the system. Under the conditions stated in Theorem 4.1.1, the output or response $y(t)$ is uniquely determined by the

(continued)

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input and the state of the system prescribed at a time t_0—that is, by the initial conditions $y(t_0), y'(t_0), \ldots, y^{(n-1)}(t_0)$.

For a dynamical system to be a linear system, it is necessary that the superposition principle (Theorem 4.1.7) holds in the system; that is, the response of the system to a superposition of inputs is a superposition of outputs. We have already examined some simple linear systems in Section 3.1 (linear first-order equations); in Section 5.1 we examine linear systems in which the mathematical models are second-order differential equations.

Exercises 4.1

Answers to selected odd-numbered problems begin on page ANS-4.

4.1.1 Initial-Value and Boundary-Value Problems

In Problems 1–4 the given family of functions is the general solution of the differential equation on the indicated interval. Find a member of the family that is a solution of the initial-value problem.

1. $y = c_1e^x + c_2e^{-x}, (-\infty, \infty)$;
   $y'' - y = 0, \quad y(0) = 0, \quad y'(0) = 1$
2. $y = c_1e^{4x} + c_2e^{-x}, (-\infty, \infty)$;
   $y'' - 3y' - 4y = 0, \quad y(0) = 1, \quad y'(0) = 2$
3. $y = c_1x + c_2x \ln x, (0, \infty)$;
   $x^2y'' - xy' + y = 0, \quad y(1) = 3, \quad y'(1) = -1$
4. $y = c_1 + c_2 \cos x + c_3 \sin x, (-\infty, \infty)$;
   $y''' + y' = 0, \quad y(\pi) = 0, \quad y'(\pi) = 2, \quad y''(\pi) = -1$
5. Given that $y = c_1 + c_2x^2$ is a two-parameter family of solutions of $xy'' - y' = 0$ on the interval $(-\infty, \infty)$, show that constants c_1 and c_2 cannot be found so that a member of the family satisfies the initial conditions $y(0) = 0, y'(0) = 1$. Explain why this does not violate Theorem 4.1.1.
6. Find two members of the family of solutions in Problem 5 that satisfy the initial conditions $y(0) = 0, y'(0) = 0$.
7. Given that $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$ is the general solution of $x'' + \omega^2x = 0$ on the interval $(-\infty, \infty)$, show that a solution satisfying the initial conditions $x(0) = x_0, x'(0) = x_1$ is given by
$$x(t) = x_0 \cos \omega t + \frac{x_1}{\omega} \sin \omega t.$$
8. Use the general solution of $x'' + \omega^2x = 0$ given in Problem 7 to show that a solution satisfying the initial conditions $x(t_0) = x_0, x'(t_0) = x_1$ is the solution given in Problem 7 shifted by an amount t_0:
$$x(t) = x_0 \cos \omega(t - t_0) + \frac{x_1}{\omega} \sin \omega(t - t_0).$$

In Problems 9 and 10 find an interval centered about $x = 0$ for which the given initial-value problem has a unique solution.

9. $(x - 2)y'' + 3y = x, \quad y(0) = 0, \quad y'(0) = 1$
10. $y'' + (\tan x)y = e^x, \quad y(0) = 1, \quad y'(0) = 0$
11. (a) Use the family in Problem 1 to find a solution of $y'' - y = 0$ that satisfies the boundary conditions $y(0) = 0, y(1) = 1$.
    (b) The DE in part (a) has the alternative general solution $y = c_3 \cosh x + c_4 \sinh x$ on $(-\infty, \infty)$. Use this family to find a solution that satisfies the boundary conditions in part (a).
    (c) Show that the solutions in parts (a) and (b) are equivalent.
12. Use the family in Problem 5 to find a solution of $xy'' - y' = 0$ that satisfies the boundary conditions $y(0) = 1, y'(1) = 6$.

In Problems 13 and 14 the given two-parameter family is a solution of the indicated differential equation on the interval $(-\infty, \infty)$. Determine whether a member of the family can be found that satisfies the boundary conditions.

13. $y = c_1e^x \cos x + c_2e^x \sin x; \quad y'' - 2y' + 2y = 0$
    (a) $y(0) = 1, \quad y'(\pi) = 0$
    (b) $y(0) = 1, \quad y(\pi) = -1$
    (c) $y(0) = 1, \quad y(\pi/2) = 1$
    (d) $y(0) = 0, \quad y(\pi) = 0.$
14. $y = c_1x^2 + c_2x^4 + 3; \quad x^2y'' - 5xy' + 8y = 24$
    (a) $y(-1) = 0, \quad y(1) = 4$
    (b) $y(0) = 1, \quad y(1) = 2$
    (c) $y(0) = 3, \quad y(1) = 0$
    (d) $y(1) = 3, \quad y(2) = 15$

4.1.2 Homogeneous Equations

In Problems 15–22 determine whether the given set of functions is linearly independent on the interval $(-\infty, \infty)$.

15. $f_1(x) = x, \quad f_2(x) = x^2, \quad f_3(x) = 4x - 3x^2$
16. $f_1(x) = 0, \quad f_2(x) = x, \quad f_3(x) = e^x$
17. $f_1(x) = 5, \quad f_2(x) = \cos^2 x, \quad f_3(x) = \sin^2 x$
18. $f_1(x) = \cos 2x, \quad f_2(x) = 1, \quad f_3(x) = \cos^2 x$

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19. $f_1(x) = x, \quad f_2(x) = x - 1, \quad f_3(x) = x + 3$

20. $f_1(x) = 2 + x, \quad f_2(x) = 2 + |x|$

21. $f_1(x) = 1 + x, \quad f_2(x) = x, \quad f_3(x) = x^2$

22. $f_1(x) = e^x, \quad f_2(x) = e^{-x}, \quad f_3(x) = \sinh x$

In Problems 23–30 verify that the given functions form a fundamental set of solutions of the differential equation on the indicated interval. Form the general solution.

23. $y'' - y' - 12y = 0; \quad e^{-3x}, e^{4x}, (-\infty, \infty)$

24. $y'' - 4y = 0; \quad \cosh 2x, \sinh 2x, (-\infty, \infty)$

25. $y'' - 2y' + 5y = 0; \quad e^x \cos 2x, e^x \sin 2x, (-\infty, \infty)$

26. $4y'' - 4y' + y = 0; \quad e^{x/2}, xe^{x/2}, (-\infty, \infty)$

27. $x^2y'' - 6xy' + 12y = 0; \quad x^3, x^4, (0, \infty)$

28. $x^2y'' + xy' + y = 0; \quad \cos(\ln x), \sin(\ln x), (0, \infty)$

29. $x^3y''' + 6x^2y'' + 4xy' - 4y = 0; \quad x, x^{-2}, x^{-2} \ln x, (0, \infty)$

30. $y^{(4)} + y'' = 0; \quad 1, x, \cos x, \sin x, (-\infty, \infty)$

4.1.3 Nonhomogeneous Equations

In Problems 31–34 verify that the given two-parameter family of functions is the general solution of the nonhomogeneous differential equation on the indicated interval.

31. $y'' - 7y' + 10y = 24e^x;$
$y = c_1e^{2x} + c_2e^{5x} + 6e^x, (-\infty, \infty)$

32. $y'' + y = \sec x;$
$y = c_1 \cos x + c_2 \sin x + x \sin x + (\cos x) \ln(\cos x), (-\pi/2, \pi/2)$

33. $y'' - 4y' + 4y = 2e^{2x} + 4x - 12;$
$y = c_1e^{2x} + c_2xe^{2x} + x^2e^{2x} + x - 2, (-\infty, \infty)$

34. $2x^2y'' + 5xy' + y = x^2 - x;$
$y = c_1x^{-1/2} + c_2x^{-1} + \frac{1}{15}x^2 - \frac{1}{6}x, (0, \infty)$

35. (a) Verify that $y_{p_1} = 3e^{2x}$ and $y_{p_2} = x^2 + 3x$ are, respectively, particular solutions of

$$y'' - 6y' + 5y = -9e^{2x}$$

and $$y'' - 6y' + 5y = 5x^2 + 3x - 16.$$

(b) Use part (a) to find particular solutions of

$$y'' - 6y' + 5y = 5x^2 + 3x - 16 - 9e^{2x}$$

and $$y'' - 6y' + 5y = -10x^2 - 6x + 32 + e^{2x}.$$

36. (a) By inspection find a particular solution of

$$y'' + 2y = 10.$$

(b) By inspection find a particular solution of

$$y'' + 2y = -4x.$$

(c) Find a particular solution of $y'' + 2y = -4x + 10$.

(d) Find a particular solution of $y'' + 2y = 8x + 5$.

Discussion Problems

37. Let $n = 1, 2, 3, \ldots$. Discuss how the observations $D^nx^{n-1} = 0$ and $D^nx^n = n!$ can be used to find the general solutions of the given differential equations.

(a) $y'' = 0$ (b) $y''' = 0$ (c) $y^{(4)} = 0$

(d) $y'' = 2$ (e) $y''' = 6$ (f) $y^{(4)} = 24$

38. Suppose that $y_1 = e^x$ and $y_2 = e^{-x}$ are two solutions of a homogeneous linear differential equation. Explain why $y_3 = \cosh x$ and $y_4 = \sinh x$ are also solutions of the equation.

39. By substituting $y = e^{mx}$ into the differential equation

$$\frac{d^4y}{dx^4} - 10\frac{d^2y}{dx^2} + 9y = 0$$

find four linearly independent solutions of the equation. Give the general solution of the equation on $(-\infty, \infty)$.

40. If y_{p_1} and y_{p_2} are particular solutions of the nonhomogeneous linear differential equation (7), then show that the difference of these particular solutions $y = y_{p_1} - y_{p_2}$ is a solution of the associated homogeneous differential equation (6).

41. (a) Verify that $y_1 = x^3$ and $y_2 = |x|^3$ are linearly independent solutions of the differential equation $x^2y'' - 4xy' + 6y = 0$ on the interval $(-\infty, \infty)$.

(b) For the functions y_1 and y_2 in part (a), show that $W(y_1, y_2) = 0$ for every real number x. Does this result violate Theorem 4.1.3? Explain.

(c) Verify that $Y_1 = x^3$ and $Y_2 = x^2$ are also linearly independent solutions of the differential equation in part (a) on the interval $(-\infty, \infty)$.

(d) Besides the functions y_1, y_2, Y_1, and Y_2 in parts (a) and (c), find a solution of the differential equation that satisfies $y(0) = 0, y'(0) = 0$.

(e) By the superposition principle, Theorem 4.1.2, both linear combinations $y = c_1y_1 + c_2y_2$ and $Y = c_1Y_1 + c_2Y_2$ are solutions of the differential equation. Discuss whether one, both, or neither of the linear combinations is a general solution of the differential equation on the interval $(-\infty, \infty)$.

42. Is the set of functions $f_1(x) = e^{x+2}, f_2(x) = e^{x-3}$ linearly dependent or linearly independent on $(-\infty, \infty)$? Discuss.

43. Suppose $y_1, y_2, \ldots, y_k$ are k linearly independent solutions on $(-\infty, \infty)$ of a homogeneous linear nth-order differential equation with constant coefficients. By Theorem 4.1.2 it follows that $y_{k+1} = 0$ is also a solution of the differential equation. Is the set of solutions $y_1, y_2, \ldots, y_k, y_{k+1}$ linearly dependent or linearly independent on $(-\infty, \infty)$? Discuss.

44. Suppose that $y_1, y_2, \ldots, y_k$ are k nontrivial solutions of a homogeneous linear nth-order differential equation with constant coefficients and that $k = n + 1$. Is the set of solutions $y_1, y_2, \ldots, y_k$ linearly dependent or linearly independent on $(-\infty, \infty)$? Discuss.

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4.2 Reduction of Order

Introduction In the preceding section we saw that the general solution of a homogeneous linear second-order differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \tag{1}$$

is a linear combination $y = c_1y_1 + c_2y_2$, where y_1 and y_2 are solutions that constitute a linearly independent set on some interval I. Beginning in the next section, we examine a method for determining these solutions when the coefficients of the differential equation in (1) are constants. This method, which is a straightforward exercise in algebra, breaks down in a few cases and yields only a single solution y_1 of the DE. It turns out that we can construct a second solution y_2 of a homogeneous equation (1) (even when the coefficients in (1) are variable) provided that we know a nontrivial solution y_1 of the DE. The basic idea described in this section is that *equation* (1) *can be reduced to a linear first-order DE by means of a substitution* involving the known solution y_1. A second solution y_2 of (1) is apparent after this first-order differential equation is solved.

Reduction of Order Suppose that y_1 denotes a nontrivial solution of (1) and that y_1 is defined on an interval I. We seek a second solution y_2 so that the set consisting of y_1 and y_2 is linearly independent on I. Recall from Section 4.1 that if y_1 and y_2 are linearly independent, then their quotient y_2/y_1 is nonconstant on I—that is, $y_2(x)/y_1(x) = u(x)$ or $y_2(x) = u(x)y_1(x)$. The function $u(x)$ can be found by substituting $y_2(x) = u(x)y_1(x)$ into the given differential equation. This method is called **reduction of order** because we must solve a linear first-order differential equation to find u.

Example 1 A Second Solution by Reduction of Order

Given that $y_1 = e^x$ is a solution of $y'' - y = 0$ on the interval $(-\infty, \infty)$, use reduction of order to find a second solution y_2.

Solution If $y = u(x)y_1(x) = u(x)e^x$, then the Product Rule gives

$$y' = ue^x + e^xu', \quad y'' = ue^x + 2e^xu' + e^xu'',$$

and so

$$y'' - y = e^x(u'' + 2u') = 0.$$

Since $e^x \neq 0$, the last equation requires $u'' + 2u' = 0$. If we make the substitution $w = u'$, this linear second-order equation in u becomes $w' + 2w = 0$, which is a linear first-order equation in w. Using the integrating factor e^{2x}, we can write $\dfrac{d}{dx}[e^{2x}w] = 0$. After integrating, we get $w = c_1e^{-2x}$ or $u' = c_1e^{-2x}$. Integrating again then yields $u = -\frac{1}{2}c_1e^{-2x} + c_2$. Thus

$$y = u(x)e^x = -\frac{c_1}{2}e^{-x} + c_2e^x. \tag{2}$$

By choosing $c_2 = 0$ and $c_1 = -2$, we obtain the desired second solution, $y_2 = e^{-x}$. Because $W(e^x, e^{-x}) \neq 0$ for every x, the solutions are linearly independent on $(-\infty, \infty)$. ■

Since we have shown that $y_1 = e^x$ and $y_2 = e^{-x}$ are linearly independent solutions of a linear second-order equation, the expression in (2) is actually the general solution of $y'' - y = 0$ on $(-\infty, \infty)$.

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General Case Suppose we divide by $a_2(x)$ to put equation (1) in the **standard form**

$$y'' + P(x)y' + Q(x)y = 0, \tag{3}$$

where $P(x)$ and $Q(x)$ are continuous on some interval I. Let us suppose further that $y_1(x)$ is a known solution of (3) on I and that $y_1(x) \neq 0$ for every x in the interval. If we define $y = u(x)y_1(x)$, it follows that

$$y' = uy_1' + y_1u', \quad y'' = uy_1'' + 2y_1'u' + y_1u''$$
$$y'' + Py' + Qy = u[\underbrace{y_1'' + Py_1' + Qy_1}_{\text{zero}}] + y_1u'' + (2y_1' + Py_1)u' = 0.$$

This implies that we must have

$$y_1u'' + (2y_1' + Py_1)u' = 0 \quad \text{or} \quad y_1w' + (2y_1' + Py_1)w = 0, \tag{4}$$

where we have let $w = u'$. Observe that the last equation in (4) is both linear and separable. Separating variables and integrating, we obtain

$$\frac{dw}{w} + 2\frac{y_1'}{y_1}dx + P\,dx = 0$$
$$\ln|wy_1^2| = -\int P\,dx + c \quad \text{or} \quad wy_1^2 = c_1e^{-\int P\,dx}.$$

We solve the last equation for w, use $w = u'$, and integrate again:

$$u = c_1\int \frac{e^{-\int P\,dx}}{y_1^2}dx + c_2.$$

By choosing $c_1 = 1$ and $c_2 = 0$, we find that $y = u(x)y_1(x)$ is a second solution of equation (3). We summarize the result in the next theorem.

Theorem 4.2.1 Reduction of Order

Let $y_1(x)$ be a solution of the homogeneous differential equation (3) on an interval I and that $y_1(x) \neq 0$ for all x in I. Then

$$y_2(x) = y_1(x)\int \frac{e^{-\int P(x)\,dx}}{y_1^2(x)}dx. \tag{5}$$

is a second solution of (3) on the interval I.

It makes a good review of differentiation to verify that the function $y_2(x)$ defined in (5) satisfies equation (3). Furthermore, if $y_2(x)$ is the solution in (5), then the functions $y_1(x)$ and $y_2(x)$ are linearly independent on any interval I on which $y_1(x)$ is not zero. To prove linear independence we use Theorem 4.1.3. The Wronskian of the functions is

$$W(y_1, y_2) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = \begin{vmatrix} y_1(x) & y_1(x)\displaystyle\int \frac{e^{-\int P(x)\,dx}}{y_1^2(x)}dx \\ y_1'(x) & \dfrac{e^{-\int P(x)\,dx}}{y_1(x)} + y_1'(x)\displaystyle\int \frac{e^{-\int P(x)\,dx}}{y_1^2(x)}dx \end{vmatrix} = e^{-\int P(x)dx} \neq 0.$$

The general solution of (3) on I is the linear combination $y(x) = c_1y_1(x) + c_2y_2(x)$.

Example 2 A Second Solution by Formula (5)

The function $y_1 = x^2$ is a solution of $x^2y'' - 3xy' + 4y = 0$. Find the general solution of the differential equation on the interval $(0, \infty)$.

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Solution From the standard form of the equation,

$$y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0,$$

we find from (5) $y_2 = x^2 \int \frac{e^{3\int dx/x}}{x^4}\,dx$ $\leftarrow e^{3\int dx/x} = e^{\ln x^3} = x^3$

$$= x^2 \int \frac{dx}{x} = x^2 \ln x.$$

The general solution on the interval $(0, \infty)$ is given by $y = c_1y_1 + c_2y_2$; that is, $y = c_1x^2 + c_2x^2 \ln x$. ■

Remarks

(i) The derivation and use of formula (5) have been illustrated here because this formula appears again in the next section and in Sections 4.7 and 6.3. We use (5) simply to save time in obtaining a desired result. Your instructor will tell you whether you should memorize (5) or whether you should know the first principles of reduction of order.

(ii) Reduction of order can be used to find the general solution of a nonhomogeneous equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x)$ whenever a solution y_1 of the associated homogeneous equation is known. See Problems 19–22 in Exercises 4.2.

(iii) The integral in (5) may be nonelementary. In this case we simply write the second solution in terms of an integral-defined function:

$$y_2(x) = y_1(x)\int_{x_0}^{x} \frac{e^{-\int P(t)\,dt}}{y_1^2(t)}\,dt,$$

where we assume that the integrand is continuous on the interval $[x_0, x]$. See Problems 23 and 24 in Exercises 4.2.

Exercises 4.2

Answers to selected odd-numbered problems begin on page ANS-4.

In Problems 1–18 the indicated function $y_1(x)$ is a solution of the given differential equation. Use reduction of order or formula (5), as instructed, to find a second solution $y_2(x)$.

1. $y'' - 4y' + 4y = 0; \quad y_1 = e^{2x}$
2. $y'' + 2y' + y = 0; \quad y_1 = xe^{-x}$
3. $y'' + 16y = 0; \quad y_1 = \cos 4x$
4. $y'' + 9y = 0; \quad y_1 = \sin 3x$
5. $y'' - y = 0; \quad y_1 = \cosh x$
6. $y'' - 25y = 0; \quad y_1 = e^{5x}$
7. $9y'' - 12y' + 4y = 0; \quad y_1 = e^{2x/3}$
8. $6y'' + y' - y = 0; \quad y_1 = e^{x/3}$
9. $x^2y'' - 7xy' + 16y = 0; \quad y_1 = x^4$
10. $x^2y'' + 2xy' - 6y = 0; \quad y_1 = x^2$
11. $xy'' + y' = 0; \quad y_1 = \ln x$
12. $4x^2y'' + y = 0; \quad y_1 = x^{1/2} \ln x$
13. $x^2y'' - xy' + 2y = 0; \quad y_1 = x \sin(\ln x)$
14. $x^2y'' - 3xy' + 5y = 0; \quad y_1 = x^2 \cos(\ln x)$
15. $(1 - 2x - x^2)y'' + 2(1 + x)y' - 2y = 0; \quad y_1 = x + 1$
16. $(1 - x^2)y'' + 2xy' = 0; \quad y_1 = 1$
17. $xy'' - (2x + 1)y' + (x + 1)y = 0; \quad y_1 = e^x$
18. $y'' + 2(\tan x)y' - y = 0; \quad y_1 = \sin x$

In Problems 19–22 the indicated function $y_1(x)$ is a solution of the associated homogeneous equation. Use the method of reduction of order to find a second solution $y_2(x)$ of the homogeneous equation and a particular solution $y_p(x)$ of the given nonhomogeneous equation.

19. $y'' - 4y = 2; \quad y_1 = e^{-2x}$
20. $y'' + y' = 1; \quad y_1 = 1$

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21. $y'' - 3y' + 2y = 5e^{3x}; \quad y_1 = e^x$

22. $y'' - 4y' + 3y = x; \quad y_1 = e^x$

In Problems 23 and 24 the indicated function $y_1(x)$ is a solution of the given differential equation. Use formula (5) to find a second solution $y_2(x)$ expressed in terms of an integral-defined function. See (*iii*) in the *Remarks*.

23. $x^2y'' + (x^2 - x)y' + (1 - x)y = 0; \quad y_1 = x$

24. $2xy'' - (2x + 1)y' + y = 0; \quad y_1 = e^x$

Discussion Problems

25. (a) Give a convincing demonstration that the second-order equation $ay'' + by' + cy = 0$, a, b, and c constants, always possesses at least one solution of the form $y_1 = e^{m_1 x}$, m_1 a constant.

(b) Explain why the differential equation in part (a) must then have a second solution either of the form $y_2 = e^{m_2 x}$ or of the form $y_2 = xe^{m_1 x}$, m_1 and m_2 constants.

(c) Reexamine Problems 1–8. Can you explain why the statements in parts (a) and (b) above are not contradicted by the answers to Problems 3–5?

26. Verify that $y_1(x) = x$ is a solution of $xy'' - xy' + y = 0$. Use reduction of order to find a second solution $y_2(x)$ in the form of an infinite series. Conjecture an interval of definition for $y_2(x)$.

Computer Lab Assignments

27. (a) Verify that $y_1(x) = e^x$ is a solution of

$$xy'' - (x + 10)y' + 10y = 0.$$

(b) Use (5) to find a second solution $y_2(x)$. Use a CAS to carry out the required integration.

(c) Explain, using Corollary (A) of Theorem 4.1.2, why the second solution can be written compactly as

$$y_2(x) = \sum_{n=0}^{10} \frac{1}{n!} x^n.$$

4.3 Homogeneous Linear Equations with Constant Coefficients

Introduction As a means of motivating the discussion in this section, let us return to first-order differential equations—more specifically, to *homogeneous* linear equations $ay' + by = 0$, where the coefficients $a \neq 0$ and b are constants. This type of equation can be solved either by separation of variables or with the aid of an integrating factor, but there is another solution method, one that uses only algebra. Before illustrating this alternative method, we make one observation: Solving $ay' + by = 0$ for y' yields $y' = ky$, where k is a constant. This observation reveals the nature of the unknown solution y; the only nontrivial elementary function whose derivative is a constant multiple of itself is an exponential function e^{mx}. Now the new solution method: If we substitute $y = e^{mx}$ and $y' = me^{mx}$ into $ay' + by = 0$, we get

$$ame^{mx} + be^{mx} = 0 \qquad \text{or} \qquad e^{mx}(am + b) = 0.$$

Since e^{mx} is never zero for real values of x, the last equation is satisfied only when m is a solution or root of the first-degree polynomial equation $am + b = 0$. For this single value of m, $y = e^{mx}$ is a solution of the DE. To illustrate, consider the constant-coefficient equation $2y' + 5y = 0$. It is not necessary to go through the differentiation and substitution of $y = e^{mx}$ into the DE; we merely have to form the equation $2m + 5 = 0$ and solve it for m. From $m = -\frac{5}{2}$ we conclude that $y = e^{-5x/2}$ is a solution of $2y' + 5y = 0$, and its general solution on the interval $(-\infty, \infty)$ is $y = c_1 e^{-5x/2}$.

In this section we will see that the foregoing procedure can produce exponential solutions for homogeneous linear higher-order DEs,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0, \tag{1}$$

where the coefficients a_i, $i = 0, 1, \ldots, n$ are real constants and $a_n \neq 0$.

Auxiliary Equation We begin by considering the special case of the homogeneous linear second-order differential equation

$$ay'' + by' + cy = 0, \tag{2}$$

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where a, b, and c are constants. If we try to find a solution of the form $y = e^{mx}$, then after substitution of $y' = me^{mx}$ and $y'' = m^2e^{mx}$, equation (2) becomes

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0 \qquad \text{or} \qquad e^{mx}(am^2 + bm + c) = 0.$$

As in the introduction we argue that because $e^{mx} \neq 0$ for all x, it is apparent that the only way $y = e^{mx}$ can satisfy the differential equation (2) is when m is chosen as a root of the quadratic equation

$$am^2 + bm + c = 0. \tag{3}$$

This last equation is called the **auxiliary equation** of the differential equation (2). Since the two roots of (3) are $m_1 = (-b + \sqrt{b^2 - 4ac})/2a$ and $m_2 = (-b - \sqrt{b^2 - 4ac})/2a$, there will be three forms of the general solution of (2) corresponding to the three cases:

- m_1 and m_2 real and distinct ($b^2 - 4ac > 0$),
- m_1 and m_2 real and equal ($b^2 - 4ac = 0$), and
- m_1 and m_2 conjugate complex numbers ($b^2 - 4ac < 0$).

We discuss each of these cases in turn.

Case I: Distinct Real Roots Under the assumption that the auxiliary equation (3) has two unequal real roots m_1 and m_2, we find two solutions, $y_1 = e^{m_1x}$ and $y_2 = e^{m_2x}$. We see that these functions are linearly independent on $(-\infty, \infty)$ and hence form a fundamental set. It follows that the general solution of (2) on this interval is

$$y = c_1e^{m_1x} + c_2e^{m_2x}. \tag{4}$$

Case II: Repeated Real Roots When $m_1 = m_2$, we necessarily obtain only one exponential solution, $y_1 = e^{m_1x}$. From the quadratic formula we find that $m_1 = -b/2a$ since the only way to have $m_1 = m_2$ is to have $b^2 - 4ac = 0$. It follows from (5) in Section 4.2 that a second solution of the equation is

$$y_2 = e^{m_1x}\int \frac{e^{2m_1x}}{e^{2m_1x}}\,dx = e^{m_1x}\int dx = xe^{m_1x}. \tag{5}$$

In (5) we have used the fact that $-b/a = 2m_1$. The general solution is then

$$y = c_1e^{m_1x} + c_2xe^{m_1x}. \tag{6}$$

Case III: Conjugate Complex Roots If m_1 and m_2 are complex, then we can write $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Formally, there is no difference between this case and Case I, and hence

$$y = C_1e^{(\alpha+i\beta)x} + C_2e^{(\alpha-i\beta)x}.$$

However, in practice we prefer to work with real functions instead of complex exponentials. To this end we use **Euler's formula:**

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

where θ is any real number.* It follows from this formula that

$$e^{i\beta x} = \cos\beta x + i\sin\beta x \qquad \text{and} \qquad e^{-i\beta x} = \cos\beta x - i\sin\beta x, \tag{7}$$

where we have used $\cos(-\beta x) = \cos\beta x$ and $\sin(-\beta x) = -\sin\beta x$. Note that by first adding and then subtracting the two equations in (7), we obtain, respectively,

$$e^{i\beta x} + e^{-i\beta x} = 2\cos\beta x \qquad \text{and} \qquad e^{i\beta x} - e^{-i\beta x} = 2i\sin\beta x.$$

*A formal derivation of Euler's formula can be obtained from the Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ by substituting $x = i\theta$, using $i^2 = -1$, $i^3 = -i$, . . . , and then separating the series into real and imaginary parts. The plausibility thus established, we can adopt $\cos\theta + i\sin\theta$ as the *definition* of $e^{i\theta}$.

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Since $y = C_1e^{(\alpha+i\beta)x} + C_2e^{(\alpha-i\beta)x}$ is a solution of (2) for any choice of the constants C_1 and C_2, the choices $C_1 = C_2 = 1$ and $C_1 = 1$, $C_2 = -1$ give, in turn, two solutions:

$$y_1 = e^{(\alpha+i\beta)x} + e^{(\alpha-i\beta)x} \quad \text{and} \quad y_2 = e^{(\alpha+i\beta)x} - e^{(\alpha-i\beta)x}.$$

But $$y_1 = e^{\alpha x}(e^{i\beta x} + e^{-i\beta x}) = 2e^{\alpha x}\cos\beta x$$

and $$y_2 = e^{\alpha x}(e^{i\beta x} - e^{-i\beta x}) = 2ie^{\alpha x}\sin\beta x.$$

Hence from Corollary (A) of Theorem 4.1.2 the last two results show that $e^{\alpha x}\cos\beta x$ and $e^{\alpha x}\sin\beta x$ are *real* solutions of (2). Moreover, these solutions form a fundamental set on $(-\infty, \infty)$. Consequently, the general solution is

$$y = c_1e^{\alpha x}\cos\beta x + c_2e^{\alpha x}\sin\beta x = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x). \tag{8}$$

Example 1 Second-Order DEs

Solve the following differential equations.

(a) $2y'' - 5y' - 3y = 0$ **(b)** $y'' - 10y' + 25y = 0$ **(c)** $y'' + 4y' + 7y = 0$

Solution We give the auxiliary equations, the roots, and the corresponding general solutions.

(a) $2m^2 - 5m - 3 = (2m + 1)(m - 3) = 0$, $m_1 = -\frac{1}{2}$, $m_2 = 3$. From (4),

$$y = c_1e^{-x/2} + c_2e^{3x}.$$

(b) $m^2 - 10m + 25 = (m - 5)^2 = 0$, $m_1 = m_2 = 5$. From (6),

$$y = c_1e^{5x} + c_2xe^{5x}.$$

(c) $m^2 + 4m + 7 = 0$, $m_1 = -2 + \sqrt{3}i$, $m_2 = -2 - \sqrt{3}i$. From (8) with $\alpha = -2$, $\beta = \sqrt{3}$,

$$y = e^{-2x}(c_1\cos\sqrt{3}x + c_2\sin\sqrt{3}x).$$ ■

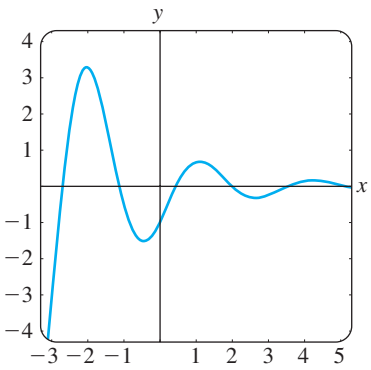


Figure 4.3.1 Solution curve of IVP in Example 2

Example 2 An Initial-Value Problem

Solve $4y'' + 4y' + 17y = 0$, $y(0) = -1$, $y'(0) = 2$.

Solution By the quadratic formula we find that the roots of the auxiliary equation $4m^2 + 4m + 17 = 0$ are $m_1 = -\frac{1}{2} + 2i$ and $m_2 = -\frac{1}{2} - 2i$. Thus from (8) we have $y = e^{-x/2}(c_1\cos 2x + c_2\sin 2x)$. Applying the condition $y(0) = -1$, we see from $e^0(c_1\cos 0 + c_2\sin 0) = -1$ that $c_1 = -1$. Differentiating $y = e^{-x/2}(-\cos 2x + c_2\sin 2x)$ and then using $y'(0) = 2$ gives $2c_2 + \frac{1}{2} = 2$ or $c_2 = \frac{3}{4}$. Hence the solution of the IVP is $y = e^{-x/2}(-\cos 2x + \frac{3}{4}\sin 2x)$. In Figure 4.3.1 we see that the solution is oscillatory, but $y \to 0$ as $x \to \infty$. ■

Two Equations Worth Knowing The two linear differential equations

$$y'' + k^2y = 0 \quad \text{and} \quad y'' - k^2y = 0,$$

where k is real, are important in applied mathematics. For $y'' + k^2y = 0$ the auxiliary equation $m^2 + k^2 = 0$ has imaginary roots $m_1 = ki$ and $m_2 = -ki$. With $\alpha = 0$ and $\beta = k$ in (8) the general solution of the DE is seen to be

$$y = c_1\cos kx + c_2\sin kx. \tag{9}$$

On the other hand, the auxiliary equation $m^2 - k^2 = 0$ for $y'' - k^2y = 0$ has distinct real roots $m_1 = k$ and $m_2 = -k$, and so by (4) the general solution of the DE is

$$y = c_1e^{kx} + c_2e^{-kx}. \tag{10}$$

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Notice that if we choose $c_1 = c_2 = \frac{1}{2}$ and $c_1 = \frac{1}{2}$, $c_2 = -\frac{1}{2}$ in (10), we get the particular solutions $y = \frac{1}{2}(e^{kx} + e^{-kx}) = \cosh kx$ and $y = \frac{1}{2}(e^{kx} - e^{-kx}) = \sinh kx$. Since $\cosh kx$ and $\sinh kx$ are linearly independent on any interval of the x-axis, an alternative form for the general solution of $y'' - k^2y = 0$ is

$$y = c_1 \cosh kx + c_2 \sinh kx. \tag{11}$$

See Problems 43 and 44 in Exercises 4.3.

Higher-Order Equations In general, to solve an nth-order differential equation (1), where the a_i, $i = 0, 1, \ldots, n$ are real constants, we must solve an nth-degree polynomial equation

$$a_n m^n + a_{n-1} m^{n-1} + \cdots + a_2 m^2 + a_1 m + a_0 = 0. \tag{12}$$

If all the roots of (12) are real and distinct, then the general solution of (1) is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}.$$

It is somewhat harder to summarize the analogues of Cases II and III because the roots of an auxiliary equation of degree greater than two can occur in many combinations. For example, a fifth-degree equation could have five distinct real roots, or three distinct real and two complex roots, or one real and four complex roots, or five real but equal roots, or five real roots but two of them equal, and so on. When m_1 is a root of multiplicity k of an nth-degree auxiliary equation (that is, k roots are equal to m_1), it can be shown that the linearly independent solutions are

$$e^{m_1 x}, \quad x e^{m_1 x}, \quad x^2 e^{m_1 x}, \ldots, \quad x^{k-1} e^{m_1 x}$$

and the general solution must contain the linear combination

$$c_1 e^{m_1 x} + c_2 x e^{m_1 x} + c_3 x^2 e^{m_1 x} + \cdots + c_k x^{k-1} e^{m_1 x}.$$

Finally, it should be remembered that when the coefficients are real, complex roots of an auxiliary equation always appear in conjugate pairs. Thus, for example, a cubic polynomial equation can have at most two complex roots.

Example 3 Third-Order DE

Solve $y''' + 3y'' - 4y = 0$.

Solution It should be apparent from inspection of $m^3 + 3m^2 - 4 = 0$ that one root is $m_1 = 1$, so $m - 1$ is a factor of $m^3 + 3m^2 - 4$. By division we find

$$m^3 + 3m^2 - 4 = (m - 1)(m^2 + 4m + 4) = (m - 1)(m + 2)^2,$$

so the other roots are $m_2 = m_3 = -2$. Thus the general solution of the DE is $y = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x}$. ■

Example 4 Fourth-Order DE

Solve $\dfrac{d^4y}{dx^4} + 2\dfrac{d^2y}{dx^2} + y = 0$.

Solution The auxiliary equation $m^4 + 2m^2 + 1 = (m^2 + 1)^2 = 0$ has roots $m_1 = m_3 = i$ and $m_2 = m_4 = -i$. Thus from Case II the solution is

$$y = C_1 e^{ix} + C_2 e^{-ix} + C_3 x e^{ix} + C_4 x e^{-ix}.$$

By Euler's formula the grouping $C_1 e^{ix} + C_2 e^{-ix}$ can be rewritten as

$$c_1 \cos x + c_2 \sin x$$

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after a relabeling of constants. Similarly, $x(C_3e^{ix} + C_4e^{-ix})$ can be expressed as $x(c_3 \cos x + c_4 \sin x)$. Hence the general solution is

$$y = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x.$$

Repeated Complex Roots Example 4 illustrates a special case when the auxiliary equation has repeated complex roots. In general, if $m_1 = \alpha + i\beta$, $\beta > 0$ is a complex root of multiplicity k of an auxiliary equation with real coefficients, then its conjugate $m_2 = \alpha - i\beta$ is also a root of multiplicity k. From the $2k$ complex-valued solutions

$$e^{(\alpha+i\beta)x}, \quad xe^{(\alpha+i\beta)x}, \quad x^2e^{(\alpha+i\beta)x}, \quad \ldots, \quad x^{k-1}e^{(\alpha+i\beta)x},$$
$$e^{(\alpha-i\beta)x}, \quad xe^{(\alpha-i\beta)x}, \quad x^2e^{(\alpha-i\beta)x}, \quad \ldots, \quad x^{k-1}e^{(\alpha-i\beta)x},$$

we conclude, with the aid of Euler's formula, that the general solution of the corresponding differential equation must then contain a linear combination of the $2k$ real linearly independent solutions

$$e^{\alpha x}\cos \beta x, \quad xe^{\alpha x}\cos \beta x, \quad x^2e^{\alpha x}\cos \beta x, \quad \ldots, \quad x^{k-1}e^{\alpha x}\cos \beta x,$$
$$e^{\alpha x}\sin \beta x, \quad xe^{\alpha x}\sin \beta x, \quad x^2e^{\alpha x}\sin \beta x, \quad \ldots, \quad x^{k-1}e^{\alpha x}\sin \beta x.$$

In Example 4 we identify $k = 2$, $\alpha = 0$, and $\beta = 1$.

Rational Roots Of course the most difficult aspect of solving constant-coefficient differential equations is finding roots of auxiliary equations of degree greater than two. Something we can try is to test the auxiliary equation for rational roots. Recall from precalculus mathematics, if $m_1 = p/q$ is a rational root (expressed in lowest terms) of a polynomial equation $a_nm^n + \cdots + a_1m + a_0 = 0$ with integer coefficients, then the integer p is a factor of the constant term a_0 and the integer q is a factor of the leading coefficient a_n.

Example 5 Finding Rational Roots

Solve $3y''' + 5y'' + 10y' - 4y = 0$.

Solution To solve the equation we must solve the cubic polynomial auxiliary equation $3m^3 + 5m^2 + 10m - 4 = 0$. With the identifications $a_0 = -4$ and $a_3 = 3$ then the integer factors of a_0 and a_3 are, respectively, p: ± 1, ± 2, ± 4 and q: ± 1, ± 3. So the possible rational roots of the cubic equation are

$$\frac{p}{q}: \pm 1, \pm 2, \pm 4, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}.$$

Each of these numbers can then be tested—say, by synthetic division. In this way we discover both the root $m_1 = \frac{1}{3}$ and the factorization

$$3m^3 + 5m^2 + 10m - 4 = \left(m - \tfrac{1}{3}\right)(3m^2 + 6m + 12).$$

The quadratic formula applied to $3m^2 + 6m + 12 = 0$ then yields the remaining two roots $m_2 = -1 - \sqrt{3}i$ and $m_3 = -1 + \sqrt{3}i$. Therefore, the general solution of the given differential equation is $y = c_1e^{x/3} + e^{-x}(c_2 \cos\sqrt{3}x + c_3 \sin\sqrt{3}x)$.

Use of Computers Finding roots or approximation of roots of auxiliary equations is a routine problem with an appropriate calculator or computer software. Polynomial equations (in one variable) of degree less than five can be solved by means of algebraic formulas using the *solve* commands in *Mathematica* and *Maple*. For auxiliary equations of degree five or greater it might be necessary to resort to numerical commands such as **NSolve** and **FindRoot** in *Mathematica*. Because of their

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capability of solving polynomial equations, it is not surprising that these computer algebra systems are also able, by means of their *dsolve* commands, to provide explicit solutions of homogeneous linear constant-coefficient differential equations.

In the classic text *Differential Equations* by Ralph Palmer Agnew* (used by the author as a student) the following statement is made:

> It is not reasonable to expect students in this course to have computing skill and equipment necessary for efficient solving of equations such as

$$4.317\frac{d^4y}{dx^4} + 2.179\frac{d^3y}{dx^3} + 1.416\frac{d^2y}{dx^2} + 1.295\frac{dy}{dx} + 3.169y = 0. \tag{13}$$

Although it is debatable whether computing skills have improved in the intervening years, it is a certainty that technology has. If one has access to a computer algebra system, equation (13) could now be considered reasonable. After simplification and some relabeling of output, *Mathematica* yields the (approximate) general solution

$$\begin{aligned} y = {} & c_1e^{-0.728852x}\cos(0.618605x) + c_2e^{-0.728852x}\sin(0.618605x) \\ & + c_3e^{0.476478x}\cos(0.759081x) + c_4e^{0.476478x}\sin(0.759081x). \end{aligned}$$

Finally, if we are faced with an initial-value problem consisting of, say, a fourth-order equation, then to fit the general solution of the DE to the four initial conditions, we must solve four linear equations in four unknowns (the c_1, c_2, c_3, c_4 in the general solution). Using a CAS to solve the system can save lots of time. See Problems 77 and 78 in Exercises 4.3 and Problem 47 in Chapter 4 in Review.

*McGraw-Hill, New York, 1960.

Exercises 4.3

Answers to selected odd-numbered problems begin on page ANS-4.

In Problems 1–14 find the general solution of the given second-order differential equation.

1. $4y'' + y' = 0$
2. $y'' - 36y = 0$
3. $y'' - y' - 6y = 0$
4. $y'' - 3y' + 2y = 0$
5. $y'' + 8y' + 16y = 0$
6. $y'' - 10y' + 25y = 0$
7. $12y'' - 5y' - 2y = 0$
8. $y'' + 4y' - y = 0$
9. $y'' + 9y = 0$
10. $3y'' + y = 0$
11. $y'' - 4y' + 5y = 0$
12. $2y'' + 2y' + y = 0$
13. $3y'' + 2y' + y = 0$
14. $2y'' - 3y' + 4y = 0$

In Problems 15–30 find the general solution of the given higher-order differential equation.

15. $y''' - 4y'' - 5y' = 0$
16. $y''' - y = 0$
17. $y''' - 5y'' + 3y' + 9y = 0$
18. $y''' + 3y'' - 4y' - 12y = 0$
19. $\dfrac{d^3u}{dt^3} + \dfrac{d^2u}{dt^2} - 2u = 0$
20. $\dfrac{d^3x}{dt^3} - \dfrac{d^2x}{dt^2} - 4x = 0$
21. $y''' + 3y'' + 3y' + y = 0$
22. $y''' - 6y'' + 12y' - 8y = 0$
23. $y^{(4)} + y''' + y'' = 0$
24. $y^{(4)} - 2y'' + y = 0$
25. $16\dfrac{d^4y}{dx^4} + 24\dfrac{d^2y}{dx^2} + 9y = 0$
26. $\dfrac{d^4y}{dx^4} - 7\dfrac{d^2y}{dx^2} - 18y = 0$
27. $\dfrac{d^5u}{dr^5} + 5\dfrac{d^4u}{dr^4} - 2\dfrac{d^3u}{dr^3} - 10\dfrac{d^2u}{dr^2} + \dfrac{du}{dr} + 5u = 0$
28. $2\dfrac{d^5x}{ds^5} - 7\dfrac{d^4x}{ds^4} + 12\dfrac{d^3x}{ds^3} + 8\dfrac{d^2x}{ds^2} = 0$
29. $\dfrac{d^6y}{dx^6} - 2\dfrac{d^5y}{dx^5} + 5\dfrac{d^4y}{dx^4} = 0$
30. $\dfrac{d^6y}{dx^6} - 3\dfrac{d^4y}{dx^4} + 3\dfrac{d^2y}{dx^2} - y = 0$

In Problems 31–38 solve the given initial-value problem.

31. $y'' + 16y = 0, \quad y(0) = 2, y'(0) = -2$
32. $\dfrac{d^2y}{d\theta^2} + y = 0, \quad y(\pi/3) = 0, y'(\pi/3) = 2$

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33. $\dfrac{d^2y}{dt^2} - 4\dfrac{dy}{dt} - 5y = 0, \quad y(1) = 0, y'(1) = 2$

34. $4y'' - 4y' - 3y = 0, \quad y(0) = 1, y'(0) = 5$

35. $y'' + y' + 2y = 0, \quad y(0) = y'(0) = 0$

36. $y'' - 2y' + y = 0, \quad y(0) = 5, y'(0) = 10$

37. $y''' + 12y'' + 36y' = 0, \quad y(0) = 0, y'(0) = 1, y''(0) = -7$

38. $y''' + 2y'' - 5y' - 6y = 0, \quad y(0) = y'(0) = 0, y''(0) = 1$

In Problems 39–42 solve the given boundary-value problem.

39. $y'' - 10y' + 25y = 0, \quad y(0) = 1, y(1) = 0$

40. $y'' + 4y = 0, \quad y(0) = 0, y(\pi) = 0$

41. $y'' + y = 0, \quad y'(0) = 0, y'(\pi/2) = 0$

42. $y'' - 2y' + 2y = 0, \quad y(0) = 1, y(\pi) = 1$

In Problems 43 and 44 solve the given problem first using the form of the general solution given in (10). Solve again, this time using the form given in (11).

43. $y'' - 3y = 0, \quad y(0) = 1, y'(0) = 5$

44. $y'' - y = 0, \quad y(0) = 1, y'(1) = 0$

In Problems 45 and 46 find all numbers $\alpha > 0$ so that the given boundary-value problem possesses nontrivial (that is, nonzero) solutions.

45. $y'' + \alpha^2 y = 0, \quad y(0) = 0, y(2) = 0$

46. $y'' + \alpha^2 y = 0, \quad y'(0) = 0, y(4) = 0$

In Problems 47–52 each figure represents the graph of a particular solution of one of the following differential equations:

(a) $y'' - 3y' - 4y = 0$ (b) $y'' + 4y = 0$

(c) $y'' + 2y' + y = 0$ (d) $y'' + y = 0$

(e) $y'' + 2y' + 2y = 0$ (f) $y'' - 3y' + 2y = 0$

Match a solution curve with one of the differential equations. Explain your reasoning.

47.

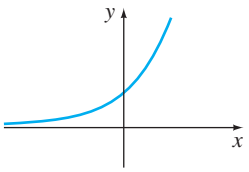


Figure 4.3.2 Graph for Problem 47

48.

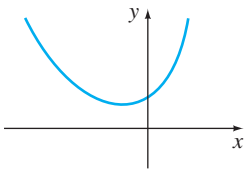


Figure 4.3.3 Graph for Problem 48

49.

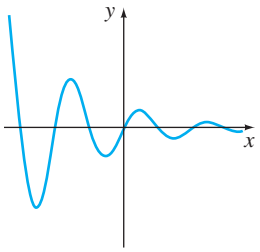


Figure 4.3.4 Graph for Problem 49

50.

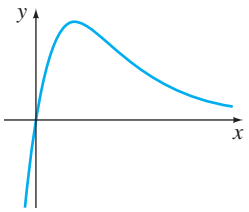


Figure 4.3.5 Graph for Problem 50

51.

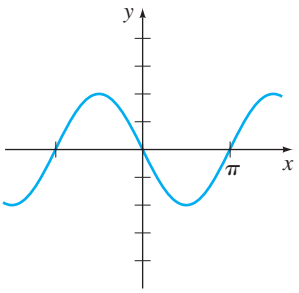


Figure 4.3.6 Graph for Problem 51

52.

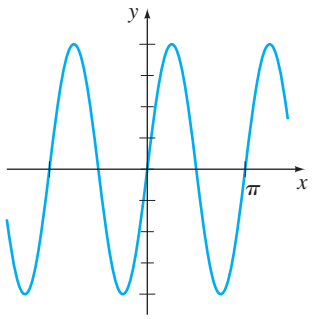


Figure 4.3.7 Graph for Problem 52

In Problems 53–62 find a homogeneous linear differential equation with constant coefficients whose general solution is given.

53. $y = c_1e^x + c_2e^{5x}$ 54. $y = c_1e^{-4x} + c_2e^{-3x}$

55. $y = c_1 + c_2e^{2x}$ 56. $y = c_1e^{10x} + c_2xe^{10x}$

57. $y = c_1 \cos 3x + c_2 \sin 3x$ 58. $y = c_1 \cosh 7x + c_2 \sinh 7x$

59. $y = c_1e^{-x}\cos x + c_2e^{-x}\sin x$

60. $y = c_1 + c_2e^{2x}\cos 5x + c_3e^{2x}\sin 5x$

61. $y = c_1 + c_2x + c_3e^{8x}$

62. $y = c_1 \cos x + c_2 \sin x + c_3 \cos 2x + c_4 \sin 2x$

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63. An equation containing a definite integral of an unknown function is naturally called an **integral equation.** An integral equation can sometimes be solved by converting it to a differential equation. Consider the integral equation

$$y(x) = 3 - 4\int_0^x y(t)dt + e^{-x}\int_0^x e^t y(t)dt.$$

(a) Use (12) of Section 1.1 to find $y'(x)$ and show that

$$e^x[y'(x) + 3y(x)] = -\int_0^x e^t y(t)dt.$$

(b) Differentiate the result in part (a) to find a homogeneous linear second-order differential equation for $y(x)$.

64. Use the first equation in Problem 63 to determine the numerical value of $y(0)$. Use the second equation in Problem 63 to determine the numerical value of $y'(0)$. Solve the differential equation found in part (b) of Problem 63 with initial conditions consisting of the values of $y(0)$ and $y'(0)$.

Discussion Problems

65. Two roots of a cubic auxiliary equation with real coefficients are $m_1 = -\frac{1}{2}$ and $m_2 = 3 + i$. What is the corresponding homogeneous linear differential equation? Discuss: Is your answer unique?

66. Find the general solution of $2y''' + 7y'' + 4y' - 4y = 0$ if $m_1 = \frac{1}{2}$ is one root of its auxiliary equation.

67. Find the general solution of $y''' + 6y'' + y' - 34y = 0$ if it is known that $y_1 = e^{-4x}\cos x$ is one solution.

68. The complex number $3 + 2i$ is a root of multiplicity 3 of the auxiliary equation of a homogeneous linear sixth-order differential equation with real constant coefficients. Find the general solution of the differential equation.

69. Find the general solution of the differential equation

$$y^{(4)} - 8y''' + 18y'' - 8y' + y = 0$$

if the auxiliary equation is

$$m^4 - 8m^3 + 18m^2 - 8m + 1 = (m^2 - 4m + 1)^2 = 0.$$

70. To solve $y^{(4)} + y = 0$, we must find the roots of $m^4 + 1 = 0$. This is a trivial problem using a CAS but can also be done by hand working with complex numbers. Observe that $m^4 + 1 = (m^2 + 1)^2 - 2m^2$. How does this help? Solve the differential equation.

71. Verify that $y = \sinh x - 2\cos(x + \pi/6)$ is a particular solution of $y^{(4)} - y = 0$. Reconcile this particular solution with the general solution of the DE.

72. Consider the boundary-value problem $y'' + \lambda y = 0$, $y(0) = 0$, $y(\pi/2) = 0$. Discuss: Is it possible to determine real values of λ so that the problem possesses **(a)** trivial solutions? **(b)** nontrivial solutions?

Computer Lab Assignments

In Problems 73–76 use a computer either as an aid in solving the auxiliary equation or as a means of directly obtaining the general solution of the given differential equation. If you use a CAS to obtain the general solution, simplify the output and, if necessary, write the solution in terms of real functions.

73. $y''' - 6y'' + 2y' + y = 0$

74. $6.11y''' + 8.59y'' + 7.93y' + 0.778y = 0$

75. $3.15y^{(4)} - 5.34y'' + 6.33y' - 2.03y = 0$

76. $y^{(4)} + 2y'' - y' + 2y = 0$

In Problems 77 and 78 use a CAS as an aid in solving the auxiliary equation. Form the general solution of the differential equation. Then use a CAS as an aid in solving the system of equations for the coefficients c_i, $i = 1, 2, 3, 4$ that results when the initial conditions are applied to the general solution.

77. $2y^{(4)} + 3y''' - 16y'' + 15y' - 4y = 0$,
$y(0) = -2, y'(0) = 6, y''(0) = 3, y'''(0) = \frac{1}{2}$

78. $y^{(4)} - 3y''' + 3y'' - y' = 0$,
$y(0) = y'(0) = 0, y''(0) = y'''(0) = 1$

4.4 Undetermined Coefficients—Superposition Approach*

Introduction To solve a nonhomogeneous linear differential equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x), \tag{1}$$

we must do two things:

- find the complementary function y_c and
- find *any* particular solution y_p of the nonhomogeneous equation (1).

*Note to the Instructor: In this section the method of undetermined coefficients is developed from the viewpoint of the superposition principle for nonhomogeneous equations (Theorem 4.1.7). In Section 4.5 an entirely different approach will be presented, one utilizing the concept of differential annihilator operators. Take your pick.

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4 Higher-Order Differential Equations

Exercises 4.1

1. From $y = c_1e^x + c_2e^{-x}$ we find $y' = c_1e^x - c_2e^{-x}$. Then $y(0) = c_1 + c_2 = 0$, $y'(0) = c_1 - c_2 = 1$ so that $c_1 = 1/2$ and $c_2 = -1/2$. The solution is $y = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$.
2. From $y = c_1e^{4x} + c_2e^{-x}$ we find $y' = 4c_1e^{4x} - c_2e^{-x}$. Then $y(0) = c_1 + c_2 = 1$, $y'(0) = 4c_1 - c_2 = 2$ so that $c_1 = 3/5$ and $c_2 = 2/5$. The solution is $y = \frac{3}{5}e^{4x} + \frac{2}{5}e^{-x}$.
3. From $y = c_1x + c_2x\ln x$ we find $y' = c_1 + c_2(1 + \ln x)$. Then $y(1) = c_1 = 3$, $y'(1) = c_1 + c_2 = -1$ so that $c_1 = 3$ and $c_2 = -4$. The solution is $y = 3x - 4x\ln x$.
4. From $y = c_1 + c_2\cos x + c_3\sin x$ we find $y' = -c_2\sin x + c_3\cos x$ and $y'' = -c_2\cos x - c_3\sin x$. Then $y(\pi) = c_1 - c_2 = 0$, $y'(\pi) = -c_3 = 2$, $y''(\pi) = c_2 = -1$ so that $c_1 = -1$, $c_2 = -1$, and $c_3 = -2$. The solution is $y = -1 - \cos x - 2\sin x$.
5. From $y = c_1 + c_2x^2$ we find $y' = 2c_2x$. Then $y(0) = c_1 = 0$, $y'(0) = 2c_2 \cdot 0 = 0$ and $y'(0) = 1$ is not possible. Since $a_2(x) = x$ is 0 at $x = 0$, Theorem 4.1 is not violated.
6. In this case we have $y(0) = c_1 = 0$, $y'(0) = 2c_2 \cdot 0 = 0$ so $c_1 = 0$ and c_2 is arbitrary. Two solutions are $y = x^2$ and $y = 2x^2$.
7. From $x(0) = x_0 = c_1$ we see that $x(t) = x_0\cos\omega t + c_2\sin\omega t$ and $x'(t) = -x_0\sin\omega t + c_2\omega\cos\omega t$. Then $x'(0) = x_1 = c_2\omega$ implies $c_2 = x_1/\omega$. Thus
$$x(t) = x_0\cos\omega t + \frac{x_1}{\omega}\sin\omega t.$$
8. Solving the system
$$x(t_0) = c_1\cos\omega t_0 + c_2\sin\omega t_0 = x$$
$$x'(t_0) = -c_1\omega\sin\omega t_0 + c_2\omega\cos\omega t_0 = x_1$$
for c_1 and c_2 gives
$$c_1 = \frac{\omega x_0\cos\omega t_0 - x_1\sin\omega t_0}{\omega} \quad \text{and} \quad c_2 = \frac{x_1\cos\omega t_0 + \omega x_0\sin\omega t_0}{\omega}.$$
Thus
$$\begin{aligned} x(t) &= \frac{\omega x_0\cos\omega t_0 - x_1\sin\omega t_0}{\omega}\cos\omega t + \frac{x_1\cos\omega t_0 + \omega x_0\sin\omega t_0}{\omega}\sin\omega t \\ &= x_0(\cos\omega t\cos\omega t_0 + \sin\omega t\sin\omega t_0) + \frac{x_1}{\omega}(\sin\omega t\cos\omega t_0 - \cos\omega t\sin\omega t_0) \\ &= x_0\cos\omega(t - t_0) + \frac{x_1}{\omega}\sin\omega(t - t_0). \end{aligned}$$

9. Since $a_2(x) = x - 2$ and $x_0 = 0$ the problem has a unique solution for $-\infty < x < 2$.

10. Since $a_0(x) = \tan x$ and $x_0 = 0$ the problem has a unique solution for $-\pi/2 < x < \pi/2$.

11. We have $y(0) = c_1 + c_2 = 0$, $y'(1) = c_1 e + c_2 e^{-1} = 1$ so that $c_1 = e/\left(e^2 - 1\right)$ and $c_2 = -e/\left(e^2 - 1\right)$. The solution is $y = e\left(e^x - e^{-x}\right)/\left(e^2 - 1\right)$.

12. In this case we have $y(0) = c_1 = 1$, $y'(1) = 2c_2 = 6$ so that $c_1 = 1$ and $c_2 = 3$. The solution is $y = 1 + 3x^2$.

13. From $y = c_1 e^x \cos x + c_2 e^x \sin x$ we find $y' = c_1 e^x(-\sin x + \cos x) + c_2 e^x(\cos x + \sin x)$.

    (a) We have $y(0) = c_1 = 1$, $y'(0) = c_1 + c_2 = 0$ so that $c_1 = 1$ and $c_2 = -1$. The solution is $y = e^x \cos x - e^x \sin x$.

    (b) We have $y(0) = c_1 = 1$, $y(\pi) = -c_1 e^\pi = -1$, which is not possible.

    (c) We have $y(0) = c_1 = 1$, $y(\pi/2) = c_2 e^{\pi/2} = 1$ so that $c_1 = 1$ and $c_2 = e^{-\pi/2}$. The solution is $y = e^x \cos x + e^{-\pi/2} e^x \sin x$.

    (d) We have $y(0) = c_1 = 0$, $y(\pi) = -c_1 e^\pi = 0$ so that $c_1 = 0$ and c_2 is arbitrary. Solutions are $y = c_2 e^x \sin x$, for any real numbers c_2.

14. (a) We have $y(-1) = c_1 + c_2 + 3 = 0$, $y(1) = c_1 + c_2 + 3 = 4$, which is not possible.

    (b) We have $y(0) = c_1 \cdot 0 + c_2 \cdot 0 + 3 = 1$, which is not possible.

    (c) We have $y(0) = c_1 \cdot 0 + c_2 \cdot 0 + 3 = 3$, $y(1) = c_1 + c_2 + 3 = 0$ so that c_1 is arbitrary and $c_2 = -3 - c_1$. Solutions are $y = c_1 x^2 - (c_1 + 3)x^4 + 3$.

    (d) We have $y(1) = c_1 + c_2 + 3 = 3$, $y(2) = 4c_1 + 16c_2 + 3 = 15$ so that $c_1 = -1$ and $c_2 = 1$. The solution is $y = -x^2 + x^4 + 3$.

15. Since $(-4)x + (3)x^2 + (1)(4x - 3x^2) = 0$ the functions are linearly dependent.

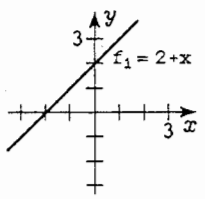
16. Since $(1)0 + (0)x + (0)e^x = 0$ the functions are linearly dependent. A similar argument shows that any set of functions containing $f(x) = 0$ will be linearly dependent.

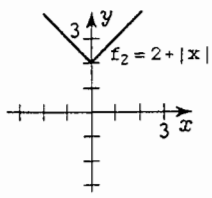
17. Since $(-1/5)5 + (1)\cos^2 x + (1)\sin^2 x = 0$ the functions are linearly dependent.

18. Since $(1)\cos 2x + (1)1 + (-2)\cos^2 x = 0$ the functions are linearly dependent.

19. Since $(-4)x + (3)(x - 1) + (1)(x + 3) = 0$ the functions are linearly dependent.

20. From the graphs of $f_1(x) = 2 + x$ and $f_2(x) = 2 + |x|$ we see that the functions are linearly independent since they cannot be multiples of each other.





21. The functions are linearly independent since $W\left(1+x, x, x^2\right) = \begin{vmatrix} 1+x & x & x^2 \\ 1 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix} = 2 \neq 0.$

22. Since $(-1/2)e^x + (1/2)e^{-x} + (1)\sinh x = 0$ the functions are linearly dependent.

23. The functions satisfy the differential equation and are linearly independent since

$$W\left(e^{-3x}, e^{4x}\right) = 7e^x \neq 0$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 e^{-3x} + c_2 e^{4x}.$$

24. The functions satisfy the differential equation and are linearly independent since

$$W(\cosh 2x, \sinh 2x) = 2$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 \cosh 2x + c_2 \sinh 2x.$$

25. The functions satisfy the differential equation and are linearly independent since

$$W\left(e^x \cos 2x, e^x \sin 2x\right) = 2e^{2x} \neq 0$$

for $-\infty < x < \infty$. The general solution is $y = c_1 e^x \cos 2x + c_2 e^x \sin 2x$.

26. The functions satisfy the differential equation and are linearly independent since

$$W\left(e^{x/2}, xe^{x/2}\right) = e^x \neq 0$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 e^{x/2} + c_2 x e^{x/2}.$$

27. The functions satisfy the differential equation and are linearly independent since

$$W\left(x^3, x^4\right) = x^6 \neq 0$$

for $0 < x < \infty$. The general solution is

$$y = c_1 x^3 + c_2 x^4.$$

28. The functions satisfy the differential equation and are linearly independent since

$$W\left(\cos(\ln x), \sin(\ln x)\right) = 1/x \neq 0$$

for $0 < x < \infty$. The general solution is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x).$$

29. The functions satisfy the differential equation and are linearly independent since

$$W\left(x, x^{-2}, x^{-2}\ln x\right) = 9x^{-6} \neq 0$$

for $0 < x < \infty$. The general solution is

$$y = c_1 x + c_2 x^{-2} + c_3 x^{-2}\ln x.$$

30. The functions satisfy the differential equation and are linearly independent since

$$W(1, x, \cos x, \sin x) = 1$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 + c_2 x + c_3 \cos x + c_4 \sin x.$$

31. The functions $y_1 = e^{2x}$ and $y_2 = e^{5x}$ form a fundamental set of solutions of the homogeneous equation, and $y_p = 6e^x$ is a particular solution of the nonhomogeneous equation.

32. The functions $y_1 = \cos x$ and $y_2 = \sin x$ form a fundamental set of solutions of the homogeneous equation, and $y_p = x\sin x + (\cos x)\ln(\cos x)$ is a particular solution of the nonhomogeneous equation.

33. The functions $y_1 = e^{2x}$ and $y_2 = xe^{2x}$ form a fundamental set of solutions of the homogeneous equation, and $y_p = x^2e^{2x} + x - 2$ is a particular solution of the nonhomogeneous equation.

34. The functions $y_1 = x^{-1/2}$ and $y_2 = x^{-1}$ form a fundamental set of solutions of the homogeneous equation, and $y_p = \frac{1}{15}x^2 - \frac{1}{6}x$ is a particular solution of the nonhomogeneous equation.

35. (a) We have $y_{p_1}' = 6e^{2x}$ and $y_{p_1}'' = 12e^{2x}$, so

$$y_{p_1}'' - 6y_{p_1}' + 5y_{p_1} = 12e^{2x} - 36e^{2x} + 15e^{2x} = -9e^{2x}.$$

Also, $y_{p_2}' = 2x + 3$ and $y_{p_2}'' = 2$, so

$$y_{p_2}'' - 6y_{p_2}' + 5y_{p_2} = 2 - 6(2x + 3) + 5(x^2 + 3x) = 5x^2 + 3x - 16.$$

(b) By the superposition principle for nonhomogeneous equations a particular solution of $y'' - 6y' + 5y = 5x^2 + 3x - 16 - 9e^{2x}$ is $y_p = x^2 + 3x + 3e^{2x}$. A particular solution of the second equation is

$$y_p = -2y_{p_2} - \frac{1}{9}y_{p_1} = -2x^2 - 6x - \frac{1}{3}e^{2x}.$$

36. (a) $y_{p_1} = 5$

(b) $y_{p_2} = -2x$

(c) $y_p = y_{p_1} + y_{p_2} = 5 - 2x$

(d) $y_p = \frac{1}{2}y_{p_1} - 2y_{p_2} = \frac{5}{2} + 4x$

37. (a) Since $D^2x = 0$, x and 1 are solutions of $y'' = 0$. Since they are linearly independent, the general solution is $y = c_1x + c_2$.

(b) Since $D^3x^2 = 0$, x^2, x, and 1 are solutions of $y''' = 0$. Since they are linearly independent, the general solution is $y = c_1x^2 + c_2x + c_3$.

(c) Since $D^4x^3 = 0$, x^3, x^2, x, and 1 are solutions of $y^{(4)} = 0$. Since they are linearly independent, the general solution is $y = c_1x^3 + c_2x^2 + c_3x + c_4$.

(d) By part (a), the general solution of $y'' = 0$ is $y_c = c_1x + c_2$. Since $D^2x^2 = 2! = 2$, $y_p = x^2$ is a particular solution of $y'' = 2$. Thus, the general solution is $y = c_1x + c_2 + x^2$.

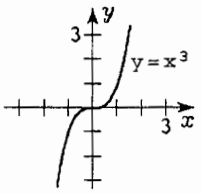
(e) By part (b), the general solution of $y''' = 0$ is $y_c = c_1x^2 + c_2x + c_3$. Since $D^3x^3 = 3! = 6$, $y_p = x^3$ is a particular solution of $y''' = 6$. Thus, the general solution is $y = c_1x^2 + c_2x + c_3 + x^3$.

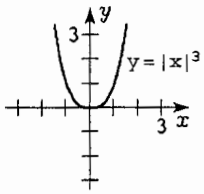
(f) By part (c), the general solution of $y^{(4)} = 0$ is $y_c = c_1x^3 + c_2x^2 + c_3x + c_4$. Since $D^4x^4 = 4! = 24$, $y_p = x^4$ is a particular solution of $y^{(4)} = 24$. Thus, the general solution is $y = c_1x^3 + c_2x^2 + c_3x + c_4 + x^4$.

38. By the superposition principle, if $y_1 = e^x$ and $y_2 = e^{-x}$ are both solutions of a homogeneous linear differential equation, then so are

$$\frac{1}{2}(y_1 + y_2) = \frac{e^x + e^{-x}}{2} = \cosh x \quad \text{and} \quad \frac{1}{2}(y_1 - y_2) = \frac{e^x - e^{-x}}{2} = \sinh x.$$

39. (a) From the graphs of $y_1 = x^3$ and $y_2 = |x|^3$ we see that the functions are linearly independent since they cannot be multiples of each other. It is easily shown that $y_1 = x^3$ solves $x^2y'' - 4xy' + 6y = 0$. To show that $y_2 = |x|^3$ is a solution let $y_2 = x^3$ for $x \geq 0$ and let $y_2 = -x^3$ for $x < 0$.





(b) If $x \geq 0$ then $y_2 = x^3$ and $W(y_1, y_2) = \begin{vmatrix} x^3 & x^3 \\ 3x^2 & 3x^2 \end{vmatrix} = 0.$

If $x < 0$ then $y_2 = -x^3$ and $W(y_1, y_2) = \begin{vmatrix} x^3 & -x^3 \\ 3x^2 & -3x^2 \end{vmatrix} = 0.$

This does not violate Theorem 4.3 since $a_2(x) = x^2$ is zero at $x = 0$.

(c) The functions $Y_1 = x^3$ and $Y_2 = x^2$ are solutions of $x^2y'' - 4xy' + 6y = 0$. They are linearly independent since $W\left(x^3, x^2\right) = x^4 \neq 0$ for $-\infty < x < \infty$.

(d) The function $y = x^3$ satisfies $y(0) = 0$ and $y'(0) = 0$.

(e) Neither is the general solution since we form a general solution on an interval for which $a_2(x) \neq 0$ for every x in the interval.

40. Since $e^{x-3} = e^{-3}e^x = (e^{-5}e^2)e^x = e^{-5}e^{x+2}$, we see that e^{x-3} is a constant multiple of e^{x+2} and the functions are linearly dependent.

41. Since $0y_1 + 0y_2 + \cdots + 0y_k + 1y_{k+1} = 0$, the set of solutions is linearly dependent.

42. The solutions are linearly dependent. Suppose n of the solutions are linearly independent (if not, then the set of $n+1$ solutions is linearly dependent). Without loss of generality, let this set be y_1, y_2, $\ldots$, y_n. Then $y = c_1y_1 + c_2y_2 + \cdots + c_ny_n$ is the general solution of the nth-order differential equation and for some choice, c_1^*, c_2^*, $\ldots$, c_n^*, of the coefficients $y_{n+1} = c_1^*y_1 + c_2^*y_2 + \cdots + c_n^*y_n$. But then the set y_1, y_2, $\ldots$, y_n, y_{n+1} is linearly dependent.

Exercises 4.2

In Problems 1-8 we use reduction of order to find a second solution. In Problems 9-16 we use formula (5) from the text.

1. Define $y = u(x)e^{2x}$ so

$$y' = 2ue^{2x} + u'e^{2x}, \quad y'' = e^{2x}u'' + 4e^{2x}u' + 4e^{2x}u, \quad \text{and} \quad y'' - 4y' + 4y = 4e^{2x}u'' = 0.$$

Therefore $u'' = 0$ and $u = c_1x + c_2$. Taking $c_1 = 1$ and $c_2 = 0$ we see that a second solution is $y_2 = xe^{2x}$.

2. Define $y = u(x)xe^{-x}$ so

$$y' = (1-x)e^{-x}u + xe^{-x}u', \quad y'' = xe^{-x}u'' + 2(1-x)e^{-x}u' - (2-x)e^{-x}u,$$

and

$$y'' + 2y' + y = e^{-x}(xu'' + 2u') = 0 \quad \text{or} \quad u'' + \frac{2}{x}u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + \dfrac{2}{x}w = 0$ which has the integrating factor $e^{2\int dx/x} = x^2$. Now

$$\frac{d}{dx}[x^2w] = 0 \quad \text{gives} \quad x^2w = c.$$

Therefore $w = u' = c/x^2$ and $u = c_1/x$. A second solution is $y_2 = \dfrac{1}{x}xe^{-x} = e^{-x}$.

3. Define $y = u(x)\cos 4x$ so

$$y' = -4u\sin 4x + u'\cos 4x, \quad y'' = u''\cos 4x - 8u'\sin 4x - 16u\cos 4x$$

and

$$y'' + 16y = (\cos 4x)u'' - 8(\sin 4x)u' = 0 \quad \text{or} \quad u'' - 8(\tan 4x)u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' - 8(\tan 4x)w = 0$ which has the integrating factor $e^{-8\int \tan 4x\,dx} = \cos^2 4x$. Now

$$\frac{d}{dx}[(\cos^2 4x)w] = 0 \quad \text{gives} \quad (\cos^2 4x)w = c.$$

Therefore $w = u' = c\sec^2 4x$ and $u = c_1 \tan 4x$. A second solution is $y_2 = \tan 4x \cos 4x = \sin 4x$.

4. Define $y = u(x)\sin 3x$ so

$$y' = 3u\cos 3x + u'\sin 3x, \quad y'' = u''\sin 3x + 6u'\cos 3x - 9u\sin 3x,$$

and

$$y'' + 9y = (\sin 3x)u'' + 6(\cos 3x)u' = 0 \quad \text{or} \quad u'' + 6(\cot 3x)u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + 6(\cot 3x)w = 0$ which has the integrating factor $e^{6\int \cot 3x\,dx} = \sin^2 3x$. Now

$$\frac{d}{dx}[(\sin^2 3x)w] = 0 \quad \text{gives} \quad (\sin^2 3x)w = c.$$

Therefore $w = u' = c\csc^2 3x$ and $u = c_1 \cot 3x$. A second solution is $y_2 = \cot 3x \sin 3x = \cos 3x$.

5. Define $y = u(x)\cosh x$ so

$$y' = u\sinh x + u'\cosh x, \quad y'' = u''\cosh x + 2u'\sinh x + u\cosh x$$

and

$$y'' - y = (\cosh x)u'' + 2(\sinh x)u' = 0 \quad \text{or} \quad u'' + 2(\tanh x)u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + 2(\tanh x)w = 0$ which has the integrating factor $e^{2\int \tanh x\,dx} = \cosh^2 x$. Now

$$\frac{d}{dx}[(\cosh^2 x)w] = 0 \quad \text{gives} \quad (\cosh^2 x)w = c.$$

Therefore $w = u' = c\,\mathrm{sech}^2 x$ and $u = c_1 \tanh x$. A second solution is $y_2 = \tanh x \cosh x = \sinh x$.

6. Define $y = u(x)e^{5x}$ so

$$y' = 5e^{5x}u + e^{5x}u', \quad y'' = e^{5x}u'' + 10e^{5x}u' + 25e^{5x}u$$

and

$$y'' - 25y = e^{5x}(u'' + 10u') = 0 \quad \text{or} \quad u'' + 10u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + 10w = 0$ which has the integrating factor $e^{10\int dx} = e^{10x}$. Now

$$\frac{d}{dx}[e^{10x}w] = 0 \quad \text{gives} \quad e^{10x}w = c.$$

Therefore $w = u' = ce^{-10x}$ and $u = c_1 e^{-10x}$. A second solution is $y_2 = e^{-10x}e^{5x} = e^{-5x}$.

7. Define $y = u(x)e^{2x/3}$ so

$$y' = \frac{2}{3}e^{2x/3}u + e^{2x/3}u', \quad y'' = e^{2x/3}u'' + \frac{4}{3}e^{2x/3}u' + \frac{4}{9}e^{2x/3}u$$

and

$$9y'' - 12y' + 4y = 9e^{2x/3}u'' = 0.$$

Therefore $u'' = 0$ and $u = c_1 x + c_2$. Taking $c_1 = 1$ and $c_2 = 0$ we see that a second solution is $y_2 = xe^{2x/3}$.

8. Define $y = u(x)e^{x/3}$ so

$$y' = \frac{1}{3}e^{x/3}u + e^{x/3}u', \quad y'' = e^{x/3}u'' + \frac{2}{3}e^{x/3}u' + \frac{1}{9}e^{x/3}u$$

and

$$6y'' + y' - y = e^{x/3}(6u'' + 5u') = 0 \quad \text{or} \quad u'' + \frac{5}{6}u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + \frac{5}{6}w = 0$ which has the integrating factor $e^{(5/6)\int dx} = e^{5x/6}$. Now

$$\frac{d}{dx}\left[e^{5x/6}w\right] = 0 \quad \text{gives} \quad e^{5x/6}w = c.$$

Therefore $w = u' = ce^{-5x/6}$ and $u = c_1 e^{-5x/6}$. A second solution is $y_2 = e^{-5x/6}e^{x/3} = e^{-x/2}$.

9. Identifying $P(x) = -7/x$ we have

$$y_2 = x^4 \int \frac{e^{-\int -(7/x)\,dx}}{x^8}\,dx = x^4 \int \frac{1}{x}\,dx = x^4 \ln|x|.$$

A second solution is $y_2 = x^4 \ln|x|$.

10. Identifying $P(x) = 2/x$ we have

$$y_2 = x^2 \int \frac{e^{-\int (2/x)\,dx}}{x^4}\,dx = x^2 \int x^{-6}dx = -\frac{1}{5}x^{-3}.$$

A second solution is $y_2 = x^{-3}$.

11. Identifying $P(x) = 1/x$ we have

$$y_2 = \ln x \int \frac{e^{-\int dx/x}}{(\ln x)^2}\,dx = \ln x \int \frac{dx}{x(\ln x)^2} = \ln x \left(-\frac{1}{\ln x}\right) = -1.$$

A second solution is $y_2 = 1$.

12. Identifying $P(x) = 0$ we have

$$y_2 = x^{1/2} \ln x \int \frac{e^{-\int 0\,dx}}{x(\ln x)^2} = x^{1/2} \ln x \left(-\frac{1}{\ln x}\right) = -x^{1/2}.$$

A second solution is $y_2 = x^{1/2}$.

13. Identifying $P(x) = -1/x$ we have

$$y_2 = x\sin(\ln x) \int \frac{e^{-\int -dx/x}}{x^2 \sin^2(\ln x)}\,dx = x\sin(\ln x) \int \frac{x}{x^2 \sin^2(\ln x)}\,dx$$

$$= [x\sin(\ln x)]\,[-\cot(\ln x)] = -x\cos(\ln x).$$

A second solution is $y_2 = x\cos(\ln x)$.

14. Identifying $P(x) = -3/x$ we have

$$y_2 = x^2\cos(\ln x)\int \frac{e^{-\int -3\,dx/x}}{x^4\cos^2(\ln x)}\,dx = x^2\cos(\ln x)\int \frac{x^3}{x^4\cos^2(\ln x)}\,dx$$

$$= x^2\cos(\ln x)\tan(\ln x) = x^2\sin(\ln x).$$

A second solution is $y_2 = x^2\sin(\ln x)$.

15. Identifying $P(x) = 2(1+x)/\left(1-2x-x^2\right)$ we have

$$y_2 = (x+1)\int \frac{e^{-\int 2(1+x)dx/\left(1-2x-x^2\right)}}{(x+1)^2}\,dx = (x+1)\int \frac{e^{\ln\left(1-2x-x^2\right)}}{(x+1)^2}\,dx$$

$$= (x+1)\int \frac{1-2x-x^2}{(x+1)^2}\,dx = (x+1)\int\left[\frac{2}{(x+1)^2}-1\right]dx$$

$$= (x+1)\left[-\frac{2}{x+1}-x\right] = -2-x^2-x.$$

A second solution is $y_2 = x^2+x+2$.

16. Identifying $P(x) = -2x/\left(1-x^2\right)$ we have

$$y_2 = \int e^{-\int -2x\,dx/(1-x^2)}dx = \int e^{-\ln(1-x^2)}dx = \int \frac{1}{1-x^2}\,dx = \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|.$$

A second solution is $y_2 = \ln|(1+x)/(1-x)|$.

17. Define $y = u(x)e^{-2x}$ so

$$y' = -2ue^{-2x} + u'e^{-2x}, \quad y'' = u''e^{-2x} - 4u'e^{-2x} + 4ue^{-2x}$$

and

$$y'' - 4y = e^{-2x}u'' - 4e^{-2x}u' = 0 \quad \text{or} \quad u'' - 4u' = 0.$$

If $w = u'$ we obtain the first order equation $w' - 4w = 0$ which has the integrating factor $e^{-4\int dx} = e^{-4x}$. Now

$$\frac{d}{dx}[e^{-4x}w] = 0 \quad \text{gives} \quad e^{-4x}w = c.$$

Therefore $w = u' = ce^{4x}$ and $u = c_1e^{4x}$. A second solution is $y_2 = e^{-2x}e^{4x} = e^{2x}$. We see by observation that a particular solution is $y_p = -1/2$. The general solution is

$$y = c_1e^{-2x} + c_2e^{2x} - \frac{1}{2}.$$

18. Define $y = u(x)\cdot 1$ so

$$y' = u', \quad y'' = u'' \quad \text{and} \quad y'' + y' = u'' + u' = 0.$$

If $w = u'$ we obtain the first order equation $w' + w = 0$ which has the integrating factor $e^{\int dx} = e^x$. Now

$$\frac{d}{dx}[e^x w] = 0 \quad \text{gives} \quad e^x w = c.$$

Therefore $w = u' = ce^{-x}$ and $u = c_1 e^{-x}$. A second solution is $y_2 = 1 \cdot e^{-x} = e^{-x}$. We see by observation that a particular solution is $y_p = x$. The general solution is

$$y = c_1 + c_2 e^{-x} + x.$$

19. Define $y = u(x)e^x$ so

$$y' = ue^x + u'e^x, \quad y'' = u''e^x + 2u'e^x + ue^x$$

and

$$y'' - 3y' + 2y = e^x u'' - e^x u' = 0 \quad \text{or} \quad u'' - u' = 0.$$

If $w = u'$ we obtain the first order equation $w' - w = 0$ which has the integrating factor $e^{-\int dx} = e^{-x}$. Now

$$\frac{d}{dx}[e^{-x} w] = 0 \quad \text{gives} \quad e^{-x} w = c.$$

Therefore $w = u' = ce^x$ and $u = ce^x$. A second solution is $y_2 = e^x e^x = e^{2x}$. To find a particular solution we try $y_p = Ae^{3x}$. Then $y' = 3Ae^{3x}$, $y'' = 9Ae^{3x}$, and $9Ae^{3x} - 3\left(3Ae^{3x}\right) + 2Ae^{3x} = 5e^{3x}$. Thus $A = 5/2$ and $y_p = \frac{5}{2}e^{3x}$. The general solution is

$$y = c_1 e^x + c_2 e^{2x} + \frac{5}{2}e^{3x}.$$

20. Define $y = u(x)e^x$ so

$$y' = ue^x + u'e^x, \quad y'' = u''e^x + 2u'e^x + ue^x$$

and

$$y'' - 4y' + 3y = e^x u'' - 2e^x u' = 0 \quad \text{or} \quad u'' - 2u' = 0.$$

If $w = u'$ we obtain the first order equation $w' - 2w = 0$ which has the integrating factor $e^{-2\int dx} = e^{-2x}$. Now

$$\frac{d}{dx}[e^{-2x} w] = 0 \quad \text{gives} \quad e^{-2x} w = c.$$

Therefore $w = u' = ce^{2x}$ and $u = c_1 e^{2x}$. A second solution is $y_2 = e^x e^{2x} = e^{3x}$. To find a particular solution we try $y_p = ax + b$. Then $y_p' = a$, $y_p'' = 0$, and $0 - 4a + 3(ax + b) = 3ax - 4a + 3b = x$. Then $3a = 1$ and $-4a + 3b = 0$ so $a = 1/3$ and $b = 4/9$. A particular solution is $y_p = \frac{1}{3}x + \frac{4}{9}$ and the general solution is

$$y = c_1 e^x + c_2 e^{3x} + \frac{1}{3}x + \frac{4}{9}.$$

Exercises 4.2

21. **(a)** For m_1 constant, let $y_1 = e^{m_1 x}$. Then $y_1' = m_1 e^{m_1 x}$ and $y_1'' = m_1^2 e^{m_1 x}$. Substituting into the differential equation we obtain

$$\begin{aligned} ay_1'' + by_1' + cy_1 &= am_1^2 e^{m_1 x} + bm_1 e^{m_1 x} + ce^{m_1 x} \\ &= e^{m_1 x}(am_1^2 + bm_1 + c) = 0. \end{aligned}$$

Thus, $y_1 = e^{m_1 x}$ will be a solution of the differential equation whenever $am_1^2 + bm_1 + c = 0$. Since a quadratic equation always has at least one real or complex root, the differential equation must have a solution of the form $y_1 = e^{m_1 x}$.

(b) Write the differential equation in the form

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = 0,$$

and let $y_1 = e^{m_1 x}$ be a solution. Then a second solution is given by

$$\begin{aligned} y_2 &= e^{m_1 x} \int \frac{e^{-bx/a}}{e^{2m_1 x}}\, dx \\ &= e^{m_1 x} \int e^{-(b/a+2m_1)x} dx \\ &= -\frac{1}{b/a + 2m_1} e^{m_1 x} e^{-(b/a+2m_1)} \qquad (m_1 \neq -b/2a) \\ &= -\frac{1}{b/a + 2m_1} e^{-(b/a+m_1)}. \end{aligned}$$

Thus, when $m_1 \neq -b/2a$, a second solution is given by $y_2 = e^{m_2 x}$ where $m_2 = -b/a - m_1$. When $m_1 = -b/2a$ a second solution is given by

$$y_2 = e^{m_1 x} \int dx = xe^{m_1 x}.$$

(c) The functions

$$\sin x = \frac{1}{2i}(e^{ix} - e^{-ix}) \qquad \cos x = \frac{1}{2}(e^{ix} + e^{-ix})$$

$$\sinh x = \frac{1}{2}(e^{x} - e^{-x}) \qquad \cosh x = \frac{1}{2}(e^{x} + e^{-x})$$

are all expressible in terms of exponential functions.

22. We have $y_1' = 1$ and $y_1'' = 0$, so $xy_1'' - xy_1' + y_1 = 0 - x + x = 0$ and $y_1(x) = x$ is a solution of the differential equation. Letting $y = u(x)y_1(x) = xu(x)$ we get

$$y' = xu'(x) + u(x) \quad \text{and} \quad y'' = xu''(x) + 2u'(x).$$

Then $xy'' - xy' + y = x^2u'' + 2xu' - x^2u' - xu + xu = x^2u'' - (x^2 - 2x)u' = 0$. If we make the substitution $w = u'$, the second-order linear differential equation becomes $x^2w' - (x^2 - x)w = 0$,

which is separable:

$$\frac{dw}{dx} = \Big(1 - \frac{1}{x}\Big)w$$

$$\frac{dw}{w} = \Big(1 - \frac{1}{x}\Big)dx$$

$$\ln w = x - \ln x + c$$

$$w = c_1\frac{e^x}{x}.$$

Then $u' = c_1e^x/x$ and $u = c_1\int e^x dx/x$. To integrate e^x/x we use the series representation for e^x. Thus, a second solution is

$$\begin{aligned} y_2 = xu(x) &= c_1x\int\frac{e^x}{x}\,dx \\ &= c_1x\int\frac{1}{x}\Big(1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots\Big)dx \\ &= c_1x\int\Big(\frac{1}{x} + 1 + \frac{1}{2!}x + \frac{1}{3!}x^2 + \cdots\Big)dx \\ &= c_1x\Big(\ln x + x + \frac{1}{2(2!)}x^2 + \frac{1}{3(3!)}x^3 + \cdots\Big) \\ &= c_1\Big(x\ln x + x^2 + \frac{1}{2(2!)}x^3 + \frac{1}{3(3!)}x^4 + \cdots\Big). \end{aligned}$$

An interval of definition is probably $(0, \infty)$ because of the $\ln x$ term.

23. **(a)** We have $y' = y'' = e^x$, so

$$xy'' - (x+10)y' + 10y = xe^x - (x+10)e^x + 10e^x = 0,$$

and $y = e^x$ is a solution of the differential equation.

(b) By (5) a second solution is

$$y_2 = y_1\int\frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx = e^x\int\frac{e^{\int\frac{x+10}{x}\,dx}}{e^{2x}}\,dx = e^x\int\frac{e^{\int(1+10/x)dx}}{e^{2x}}\,dx$$

$$= e^x\int\frac{e^{x+\ln x^{10}}}{e^{2x}}\,dx = e^x\int x^{10}e^{-x}\,dx$$

$$= e^x(-3{,}628{,}800 - 3{,}628{,}800x - 1{,}814{,}400x^2 - 604{,}800x^3 - 151{,}200x^4$$
$$- 30{,}240x^5 - 5{,}040x^6 - 720x^7 - 90x^8 - 10x^9 - x^{10})e^{-x}$$

$$= -3{,}628{,}800 - 3{,}628{,}800x - 1{,}814{,}400x^2 - 604{,}800x^3 - 151{,}200x^4$$
$$- 30{,}240x^5 - 5{,}040x^6 - 720x^7 - 90x^8 - 10x^9 - x^{10}.$$

(c) By Corollary (A) of Theorem 4.2, $-\dfrac{1}{10!}y_2 = \displaystyle\sum_{n=0}^{10}\frac{1}{n!}x^n$ is a solution.

Exercises 4.3

1. From $4m^2 + m = 0$ we obtain $m = 0$ and $m = -1/4$ so that $y = c_1 + c_2e^{-x/4}$.
2. From $m^2 - 36 = 0$ we obtain $m = 6$ and $m = -6$ so that $y = c_1e^{6x} + c_2e^{-6x}$.
3. From $m^2 - m - 6 = 0$ we obtain $m = 3$ and $m = -2$ so that $y = c_1e^{3x} + c_2e^{-2x}$.
4. From $m^2 - 3m + 2 = 0$ we obtain $m = 1$ and $m = 2$ so that $y = c_1e^{x} + c_2e^{2x}$.
5. From $m^2 + 8m + 16 = 0$ we obtain $m = -4$ and $m = -4$ so that $y = c_1e^{-4x} + c_2xe^{-4x}$.
6. From $m^2 - 10m + 25 = 0$ we obtain $m = 5$ and $m = 5$ so that $y = c_1e^{5x} + c_2xe^{5x}$.
7. From $12m^2 - 5m - 2 = 0$ we obtain $m = -1/4$ and $m = 2/3$ so that $y = c_1e^{-x/4} + c_2e^{2x/3}$.
8. From $m^2 + 4m - 1 = 0$ we obtain $m = -2 \pm \sqrt{5}$ so that $y = c_1e^{(-2+\sqrt{5})x} + c_2e^{(-2-\sqrt{5})x}$.
9. From $m^2 + 9 = 0$ we obtain $m = 3i$ and $m = -3i$ so that $y = c_1\cos 3x + c_2\sin 3x$.
10. From $3m^2 + 1 = 0$ we obtain $m = i/\sqrt{3}$ and $m = -i/\sqrt{3}$ so that $y = c_1\cos x/\sqrt{3} + c_2\sin x/\sqrt{3}$.
11. From $m^2 - 4m + 5 = 0$ we obtain $m = 2 \pm i$ so that $y = e^{2x}(c_1\cos x + c_2\sin x)$.
12. From $2m^2 + 2m + 1 = 0$ we obtain $m = -1/2 \pm i/2$ so that
$$y = e^{-x/2}(c_1\cos x/2 + c_2\sin x/2).$$
13. From $3m^2 + 2m + 1 = 0$ we obtain $m = -1/3 \pm \sqrt{2}\,i/3$ so that
$$y = e^{-x/3}\left(c_1\cos\sqrt{2}\,x/3 + c_2\sin\sqrt{2}\,x/3\right).$$
14. From $2m^2 - 3m + 4 = 0$ we obtain $m = 3/4 \pm \sqrt{23}\,i/4$ so that
$$y = e^{3x/4}\left(c_1\cos\sqrt{23}\,x/4 + c_2\sin\sqrt{23}\,x/4\right).$$
15. From $m^3 - 4m^2 - 5m = 0$ we obtain $m = 0$, $m = 5$, and $m = -1$ so that
$$y = c_1 + c_2e^{5x} + c_3e^{-x}.$$
16. From $m^3 - 1 = 0$ we obtain $m = 1$ and $m = -1/2 \pm \sqrt{3}\,i/2$ so that
$$y = c_1e^{x} + e^{-x/2}\left(c_2\cos\sqrt{3}\,x/2 + c_3\sin\sqrt{3}\,x/2\right).$$
17. From $m^3 - 5m^2 + 3m + 9 = 0$ we obtain $m = -1$, $m = 3$, and $m = 3$ so that
$$y = c_1e^{-x} + c_2e^{3x} + c_3xe^{3x}.$$

18. From $m^3 + 3m^2 - 4m - 12 = 0$ we obtain $m = -2$, $m = 2$, and $m = -3$ so that

$$y = c_1 e^{-2x} + c_2 e^{2x} + c_3 e^{-3x}.$$

19. From $m^3 + m^2 - 2 = 0$ we obtain $m = 1$ and $m = -1 \pm i$ so that

$$u = c_1 e^{t} + e^{-t}(c_2 \cos t + c_3 \sin t).$$

20. From $m^3 - m^2 - 4 = 0$ we obtain $m = 2$ and $m = -1/2 \pm \sqrt{7}\, i/2$ so that

$$x = c_1 e^{2t} + e^{-t/2}\left(c_2 \cos \sqrt{7}\, t/2 + c_3 \sin \sqrt{7}\, t/2\right).$$

21. From $m^3 + 3m^2 + 3m + 1 = 0$ we obtain $m = -1$, $m = -1$, and $m = -1$ so that

$$y = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x}.$$

22. From $m^3 - 6m^2 + 12m - 8 = 0$ we obtain $m = 2$, $m = 2$, and $m = 2$ so that

$$y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}.$$

23. From $m^4 + m^3 + m^2 = 0$ we obtain $m = 0$, $m = 0$, and $m = -1/2 \pm \sqrt{3}\, i/2$ so that

$$y = c_1 + c_2 x + e^{-x/2}\left(c_3 \cos \sqrt{3}\, x/2 + c_4 \sin \sqrt{3}\, x/2\right).$$

24. From $m^4 - 2m^2 + 1 = 0$ we obtain $m = 1$, $m = 1$, $m = -1$, and $m = -1$ so that

$$y = c_1 e^{x} + c_2 x e^{x} + c_3 e^{-x} + c_4 x e^{-x}.$$

25. From $16m^4 + 24m^2 + 9 = 0$ we obtain $m = \pm\sqrt{3}\, i/2$ and $m = \pm\sqrt{3}\, i/2$ so that

$$y = c_1 \cos \sqrt{3}\, x/2 + c_2 \sin \sqrt{3}\, x/2 + c_3 x \cos \sqrt{3}\, x/2 + c_4 x \sin \sqrt{3}\, x/2.$$

26. From $m^4 - 7m^2 - 18 = 0$ we obtain $m = 3$, $m = -3$, and $m = \pm\sqrt{2}\, i$ so that

$$y = c_1 e^{3x} + c_2 e^{-3x} + c_3 \cos \sqrt{2}\, x + c_4 \sin \sqrt{2}\, x.$$

27. From $m^5 + 5m^4 - 2m^3 - 10m^2 + m + 5 = 0$ we obtain $m = -1$, $m = -1$, $m = 1$, and $m = 1$, and $m = -5$ so that

$$u = c_1 e^{-r} + c_2 r e^{-r} + c_3 e^{r} + c_4 r e^{r} + c_5 e^{-5r}.$$

28. From $2m^5 - 7m^4 + 12m^3 + 8m^2 = 0$ we obtain $m = 0$, $m = 0$, $m = -1/2$, and $m = 2 \pm 2i$ so that

$$x = c_1 + c_2 s + c_3 e^{-s/2} + e^{2s}(c_4 \cos 2s + c_5 \sin 2s).$$

29. From $m^2 + 16 = 0$ we obtain $m = \pm 4i$ so that $y = c_1 \cos 4x + c_2 \sin 4x$. If $y(0) = 2$ and $y'(0) = -2$ then $c_1 = 2$, $c_2 = -1/2$, and $y = 2\cos 4x - \frac{1}{2}\sin 4x$.

30. From $m^2 + 1 = 0$ we obtain $m = \pm i$ so that $y = c_1 \cos\theta + c_2 \sin\theta$. If $y(\pi/3) = 0$ and $y'(\pi/3) = 2$ then $\dfrac{1}{2}c_1 + \dfrac{\sqrt{3}}{2}c_2 = 0$, $-\dfrac{\sqrt{3}}{2}c_1 + \dfrac{1}{2}c_2 = 2$, so $c_1 = -\sqrt{3}$, $c_2 = 1$, and $y = -\sqrt{3}\cos\theta + \sin\theta$.

Exercises 4.3

31. From $m^2 - 4m - 5 = 0$ we obtain $m = -1$ and $m = 5$, so that $y = c_1e^{-x} + c_2e^{5x}$. If $y(1) = 0$ and $y'(1) = 2$, then $c_1e^{-1} + c_2e^5 = 0$, $-c_1e^{-1} + 5c_2e^5 = 2$, so $c_1 = -e/3$, $c_2 = e^{-5}/3$, and $y = -\frac{1}{3}e^{1-x} + \frac{1}{3}e^{5x-5}$.

32. From $4m^2-4m-3 = 0$ we obtain $m = -1/2$ and $m = 3/2$ so that $y = c_1e^{-x/2}+c_2e^{3x/2}$. If $y(0) = 1$ and $y'(0) = 5$ then $c_1+c_2 = 1$, $-\frac{1}{2}c_1+\frac{3}{2}c_2 = 5$, so $c_1 = -7/4$, $c_2 = 11/4$, and $y = -\frac{7}{4}e^{-x/2}+\frac{11}{4}e^{3x/2}$.

33. From $m^2+m+2 = 0$ we obtain $m = -1/2\pm\sqrt{7}\,i/2$ so that $y = e^{-x/2}\left(c_1\cos\sqrt{7}\,x/2 + c_2\sin\sqrt{7}\,x/2\right)$. If $y(0) = 0$ and $y'(0) = 0$ then $c_1 = 0$ and $c_2 = 0$ so that $y = 0$.

34. From $m^2 - 2m + 1 = 0$ we obtain $m = 1$ and $m = 1$ so that $y = c_1e^x + c_2xe^x$. If $y(0) = 5$ and $y'(0) = 10$ then $c_1 = 5$, $c_1 + c_2 = 10$ so $c_1 = 5$, $c_2 = 5$, and $y = 5e^x + 5xe^x$.

35. From $m^3+12m^2+36m = 0$ we obtain $m = 0$, $m = -6$, and $m = -6$ so that $y = c_1+c_2e^{-6x}+c_3xe^{-6x}$. If $y(0) = 0$, $y'(0) = 1$, and $y''(0) = -7$ then

$$c_1 + c_2 = 0, \quad -6c_2 + c_3 = 1, \quad 36c_2 - 12c_3 = -7,$$

so $c_1 = 5/36$, $c_2 = -5/36$, $c_3 = 1/6$, and $y = \frac{5}{36} - \frac{5}{36}e^{-6x} + \frac{1}{6}xe^{-6x}$.

36. From $m^3 + 2m^2 - 5m - 6 = 0$ we obtain $m = -1$, $m = 2$, and $m = -3$ so that

$$y = c_1e^{-x} + c_2e^{2x} + c_3e^{-3x}.$$

If $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$ then

$$c_1 + c_2 + c_3 = 0, \quad -c_1 + 2c_2 - 3c_3 = 0, \quad c_1 + 4c_2 + 9c_3 = 1,$$

so $c_1 = -1/6$, $c_2 = 1/15$, $c_3 = 1/10$, and

$$y = -\frac{1}{6}e^{-x} + \frac{1}{15}e^{2x} + \frac{1}{10}e^{-3x}.$$

37. From $m^2 - 10m + 25 = 0$ we obtain $m = 5$ and $m = 5$ so that $y = c_1e^{5x} + c_2xe^{5x}$. If $y(0) = 1$ and $y(1) = 0$ then $c_1 = 1$, $c_1e^5 + c_2e^5 = 0$, so $c_1 = 1$, $c_2 = -1$, and $y = e^{5x} - xe^{5x}$.

38. From $m^2 + 4 = 0$ we obtain $m = \pm 2i$ so that $y = c_1\cos 2x + c_2\sin 2x$. If $y(0) = 0$ and $y(\pi) = 0$ then $c_1 = 0$ and $y = c_2\sin 2x$.

39. From $m^2 + 1 = 0$ we obtain $m = \pm i$ so that $y = c_1\cos x + c_2\sin x$. If $y'(0) = 0$ and $y'(\pi/2) = 2$ then $c_1 = -2$, $c_2 = 0$, and $y = -2\cos x$.

40. From $m^2 - 2m + 2 = 0$ we obtain $m = 1 \pm i$ so that $y = e^x(c_1\cos x + c_2\sin x)$. If $y(0) = 1$ and $y(\pi) = 1$ then $c_1 = 1$ and $y(\pi) = e^\pi\cos\pi = -e^\pi$. Since $-e^\pi \neq 1$, the boundary-value problem has no solution.

41. The auxiliary equation is $m^2 - 3 = 0$ which has roots $-\sqrt{3}$ and $\sqrt{3}$. By (10) the general solution is $y = c_1e^{\sqrt{3}x} + c_2e^{-\sqrt{3}x}$. By (11) the general solution is $y = c_1\cosh\sqrt{3}x + c_2\sinh\sqrt{3}x$. For $y = c_1e^{\sqrt{3}x} + c_2e^{-\sqrt{3}x}$ the initial conditions imply $c_1 + c_2 = 1$, $\sqrt{3}c_1 - \sqrt{3}c_2 = 5$. Solving for $c_1 + c_2$

we find $c_1 = \frac{1}{6}(3+5\sqrt{3})$ and $c_2 = \frac{1}{6}(3-5\sqrt{3})$ so $y = \frac{1}{6}(3+5\sqrt{3})e^{\sqrt{3}x} + \frac{1}{6}(3-5\sqrt{3})e^{\sqrt{3}x}$. For $y = c_1\cosh\sqrt{3}x + c_2\sinh\sqrt{3}x$ the initial conditions imply $c_1 = 1$, $\sqrt{3}c_2 = 5$. Solving for c_1 and c_2 we find $c_1 = 1$ and $c_2 = \frac{5}{3}\sqrt{3}$ so $y = \cosh\sqrt{3}x + \frac{5}{3}\sqrt{3}\sinh\sqrt{3}x$.

42. The auxiliary equation is $m^2 - 1 = 0$ which has roots -1 and 1. By (10) the general solution is $y = c_1e^x + c_2e^{-x}$. By (11) the general solution is $y = c_1\cosh x + c_2\sinh x$. For $y = c_1e^x + c_2e^{-x}$ the boundary conditions imply $c_1 + c_2 = 1$, $c_1e - c_2e^{-1} = 0$. Solving for c_1 and c_2 we find $c_1 = 1/(1+e^2)$ and $c_2 = e^2/(1+e^2)$ so $y = e^x/(1+e^2) + e^2e^{-x}/(1+e^2)$. For $y = c_1\cosh x + c_2\sinh x$ the boundary conditions imply $c_1 = 1$, $c_2 = -\tanh 1$, so $y = \cosh x - (\tanh 1)\sinh x$.

43. The auxiliary equation should have two positive roots, so that the solution has the form $y = c_1e^{k_1x} + c_2e^{k_2x}$. Thus, the differential equation is (f).

44. The auxiliary equation should have one positive and one negative root, so that the solution has the form $y = c_1e^{k_1x} + c_2e^{-k_2x}$. Thus, the differential equation is (a).

45. The auxiliary equation should have a pair of complex roots $a \pm bi$ where $a < 0$, so that the solution has the form $e^{ax}(c_1\cos bx + c_2\sin bx)$. Thus, the differential equation is (e).

46. The auxiliary equation should have a repeated negative root, so that the solution has the form $y = c_1e^{-x} + c_2xe^{-x}$. Thus, the differential equation is (c).

47. The differential equation should have the form $y'' + k^2y = 0$ where $k = 1$ so that the period of the solution is 2π. Thus, the differential equation is (d).

48. The differential equation should have the form $y'' + k^2y = 0$ where $k = 2$ so that the period of the solution is π. Thus, the differential equation is (b).

49. **(a)** The auxiliary equation is $m^2 - 64/L = 0$ which has roots $\pm 8/\sqrt{L}$. Thus, the general solution of the differential equation is $x = c_1\cosh(8t/\sqrt{L}) + c_2\sinh(8t/\sqrt{L})$.

(b) Setting $x(0) = x_0$ and $x'(0) = 0$ we have $c_1 = x_0$, $8c_2/\sqrt{L} = 0$. Solving for c_1 and c_2 we get $c_1 = x_0$ and $c_2 = 0$, so $x(t) = x_0\cosh(8t/\sqrt{L})$.

(c) When $L = 20$ and $x_0 = 1$, $x(t) = \cosh(4t\sqrt{5})$. The chain will last touch the peg when $x(t) = 10$. Solving $x(t) = 10$ for t we get $t_1 = \frac{1}{4}\sqrt{5}\cosh^{-1}10 \approx 1.67326$. The velocity of the chain at this instant is $x'(t_1) = 12\sqrt{11/5} \approx 17.7989$ ft/s.

50. Both $-C[1]$ and c_1 represent arbitrary constants, and each may take on any real value.

51. Since $(m-4)(m+5)^2 = m^3+6m^2-15m-100$ the differential equation is $y''' + 6y'' - 15y' - 100y = 0$. The differential equation is not unique since any constant multiple of the left-hand side of the differential equation would lead to the auxiliary roots.

52. A third root must be $m_3 = 3 - i$ and the auxiliary equation is

$$\left(m+\frac{1}{2}\right)[m-(3+i)][m-(3-i)] = \left(m+\frac{1}{2}\right)(m^2-6x+10) = m^3 - \frac{11}{2}m^2 + 7m + 5.$$

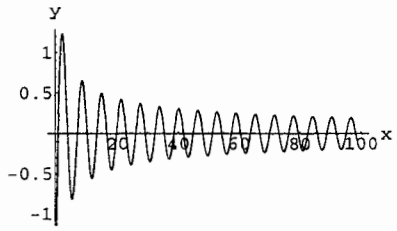
The differential equation is

$$y''' - \frac{11}{2}y'' + 7y' + 5y = 0.$$

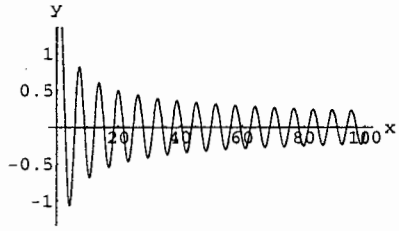
53. From the solution $y_1 = e^{-4x}\cos x$ we conclude that $m_1 = -4 + i$ and $m_2 = -4 - i$ are roots of the auxiliary equation. Hence another solution must be $y_2 = e^{-4x}\sin x$. Now dividing the polynomial $m^3 + 6m^2 + m - 34$ by $[m-(-4+i)][m-(-4-i)] = m^2 + 8m + 17$ gives $m - 2$. Therefore $m_3 = 2$ is the third root of the auxiliary equation, and the general solution of the differential equation is

$$y = c_1 e^{-4x}\cos x + c_2 e^{-4x}\sin x + c_3 e^{2x}.$$

54. Since $1/x \to 0$ as $x \to \infty$, we would expect the solutions of $y'' + (1/x)y' + y = 0$ to behave similar to the solutions of $y'' + y = 0$; that is, like $\sin x$ and $\cos x$ for large values of x. Solutions of $xy'' + y' + xy = 0$ are obtained using an ODE solver and are shown below with the indicated initial conditions.



$y(1) = 0, \quad y'(1) = 2$



$y(1) = 2, \quad y'(1) = 0$

55. Factoring the difference of two squares we obtain

$$m^4 + 1 = (m^2+1)^2 - 2m^2 = (m^2 + 1 - \sqrt{2}\,m)(m^2 + 1 + \sqrt{2}\,m) = 0.$$

Using the quadratic formula on each factor we get $m = \pm\sqrt{2}/2 \pm \sqrt{2}\,i/2$. The solution of the differential equation is

$$y(x) = e^{\sqrt{2}x/2}\left(c_1\cos\frac{\sqrt{2}}{2}x + c_2\sin\frac{\sqrt{2}}{2}x\right) + e^{-\sqrt{2}x/2}\left(c_3\cos\frac{\sqrt{2}}{2}x + c_4\sin\frac{\sqrt{2}}{2}x\right).$$

56. (a) The auxiliary equation $m^2 + bm + c = 0$ has solutions $m = (-b \pm \sqrt{b^2 - 4c})/2$. If $b \le 0$, then the solution will contain a term of the form $e^{\beta x}$ for $\beta > 0$, and the solution cannot approach 0 as $x \to \infty$. Thus, for the solution to approach 0 we must have $b > 0$. Now, if $c < 0$ then $\sqrt{b^2 - 4c} > b$ and $-b + \sqrt{b^2 - 4c} > 0$. Thus $y(x)$ cannot approach 0. Finally, if $c > 0$ then $\sqrt{b^2 - 4c} < b$ and $-b \pm \sqrt{b^2 - 4c} < 0$. In this case the solution has terms of the form $e^{\beta x}$ where $\beta < 0$. Therefore $y(x) \to 0$ as $x \to \infty$ if and only if $b > 0$ and $c > 0$.

(b) If $b^2 - 4c > 0$, then $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ and the only solution satisfying $y(0) = 0$, $y(1) = 0$ is $y = 0$.

If $b^2 - 4c = 0$, then $y = c_1 e^{mx} + c_2 t e^{mx}$. Again, the only solution satisfying $y(0) = 0$, $y(1) = 0$ is $y = 0$.

If $b^2 - 4c < 0$ then

$$y = c_1 e^{-bx/2} \cos \sqrt{4c - b^2}\, x + c_2 e^{-bx/2} \sin \sqrt{4c - b^2}\, x.$$

Now $y(0) = 0$ implies $c_1 = 0$ and

$$y = c_2 e^{-bx/2} \sin \sqrt{4c - b^2}\, x.$$

If we are to have a nontrivial solution, the condition $y(1) = 0$ implies $\sqrt{4c - b^2} = n\pi$ or $4c - b^2 = n^2\pi^2$ for n a positive integer.

57. The auxiliary equation is $m^2 + \lambda = 0$ and we consider three cases.

Case I When $\lambda = 0$ the general solution of the differential equation is $y = c_1 + c_2 x$. The boundary conditions imply $0 = y(0) = c_1$ and $0 = y(\pi/2) = c_2\pi/2$, so that $c_1 = c_2 = 0$ and the problem possesses only the trivial solution.

Case II When $\lambda < 0$ the general solution of the differential equation is $y = c_1 e^{\sqrt{-\lambda}\,x} + c_2 e^{-\sqrt{-\lambda}\,x}$, or alternatively, $y = c_1 \cosh \sqrt{-\lambda}\,x + c_2 \sinh \sqrt{-\lambda}\,x$. Again, $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 \sinh \sqrt{-\lambda}\,x$. The second boundary condition implies $0 = y(\pi/2) = c_2 \sinh \sqrt{-\lambda}\,\pi/2$ or $c_2 = 0$. In this case also, the problem possesses only the trivial solution.

Case III When $\lambda > 0$ the general solution of the differential equation is $y = c_1 \cos \sqrt{\lambda}\,x + c_2 \sin \sqrt{\lambda}\,x$. In this case also, $y(0) = 0$ yields $c_1 = 0$, so that $y = c_2 \sin \sqrt{\lambda}\,x$. The second boundary condition implies $0 = c_2 \sin \sqrt{\lambda}\,\pi/2$. When $\sqrt{\lambda}\,\pi/2$ is an integer multiple of π, that is, when $\sqrt{\lambda} = 2k$ for k a nonzero integer, the problem will have nontrivial solutions. Thus, for $\lambda = 4k^2$ the boundary-value problem will have nontrivial solutions $y = c_2 \sin 2kx$, where k is a nonzero integer. On the other hand, when $\sqrt{\lambda}$ is not an even integer, the boundary-value problem will have only the trivial solution.

58. Applying integration by parts twice we have

$$\begin{aligned}
\int e^{ax} f(x)\, dx &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a} \int e^{ax} f'(x)\, dx \\
&= \frac{1}{a} e^{ax} f(x) - \frac{1}{a} \left[\frac{1}{a} e^{ax} f'(x) - \frac{1}{a} \int e^{ax} f''(x)\, dx \right] \\
&= \frac{1}{a} e^{ax} f(x) - \frac{1}{a^2} e^{ax} f'(x) + \frac{1}{a^2} \int e^{ax} f''(x)\, dx.
\end{aligned}$$

Collecting the integrals we get

$$\int e^{ax} \left(f(x) - \frac{1}{a^2} f''(x) \right) dx = \frac{1}{a} e^{ax} f(x) - \frac{1}{a^2} e^{ax} f'(x).$$

In order for the technique to work we need to have

$$\int e^{ax}\left(f(x)-\frac{1}{a^2}f''(x)\right)dx = k\int e^{ax}f(x)\,dx$$

or

$$f(x)-\frac{1}{a^2}f''(x) = kf(x),$$

where $k \neq 0$. This is the second-order differential equation

$$f''(x)+a^2(k-1)f(x) = 0.$$

If $k < 1$, $k \neq 0$, the solution of the differential equation is a pair of exponential functions, in which case the original integrand is an exponential function and does not require integration by parts for its evaluation. Similarly, if $k = 1$, $f''(x) = 0$ and $f(x)$ has the form $f(x) = ax + b$. In this case a single application of integration by parts will suffice. Finally, if $k > 1$, the solution of the differential equation is

$$f(x) = c_1\cos a\sqrt{k-1}\,x + c_2\sin a\sqrt{k-1}\,x,$$

and we see that the technique will work for linear combinations of $\cos\alpha x$ and $\sin\alpha x$.

59. Using a CAS to solve the auxiliary equation $m^3 - 6m^2 + 2m + 1$ we find $m_1 = -0.270534$, $m_2 = 0.658675$, and $m_3 = 5.61186$. The general solution is

$$y = c_1e^{-0.270534x} + c_2e^{0.658675x} + c_3e^{5.61186x}.$$

60. Using a CAS to solve the auxiliary equation $6.11m^3 + 8.59m^2 + 7.93m + 0.778 = 0$ we find $m_1 = -0.110241$, $m_2 = -0.647826 + 0.857532i$, and $m_3 = -0.647826 - 0.857532i$. The general solution is

$$y = c_1e^{-0.110241x} + e^{-0.647826x}(c_2\cos 0.857532x + c_3\sin 0.857532x).$$

61. Using a CAS to solve the auxiliary equation $3.15m^4 - 5.34m^2 + 6.33m - 2.03 = 0$ we find $m_1 = -1.74806$, $m_2 = 0.501219$, $m_3 = 0.62342 + 0.588965i$, and $m_4 = 0.62342 - 0.588965i$. The general solution is

$$y = c_1e^{-1.74806x} + c_2e^{0.501219x} + e^{0.62342x}(c_3\cos 0.588965x + c_4\sin 0.588965x).$$

62. Using a CAS to solve the auxiliary equation $m^4 + 2m^2 - m + 2 = 0$ we find $m_1 = 1/2 + \sqrt{3}\,i/2$, $m_2 = 1/2 - \sqrt{3}\,i/2$, $m_3 = -1/2 + \sqrt{7}\,i/2$, and $m_4 = -1/2 - \sqrt{7}\,i/2$. The general solution is

$$y = e^{x/2}\left(c_1\cos\frac{\sqrt{3}}{2}x + c_2\sin\frac{\sqrt{3}}{2}x\right) + e^{-x/2}\left(c_3\cos\frac{\sqrt{7}}{2}x + c_4\sin\frac{\sqrt{7}}{2}x\right).$$